

ML1: Introduction to Machine Learning for Everybody

SE GP G QPI QM P

Training Objective

By the end of this course, you will ...

- Have a better understanding of what machine learning is all about
- Understand some basic principals of machine learning
- Have made **first experiences** with running machine learning algorithms yourself
- Discuss application opportunities of Machine Learning in our business
- Tailoring Knowledge to your needs in our Energy business



This training is an experiment

- New teaching approach targeting “everybody”
- New teaching over Circuit/ pure recordings
- We are using EnergyAI, Python and Jupiter Notebooks in this course. We explain more about these in our Data Science trainings. <https://wse10.siemens-energy.com/content/P0002338/SitePages/Data%20Science%20Training%20and%20Multiplier%20Networking.aspx>
- However, we don’t want to make them a prerequisite, so that everybody can attend.

How this Training Works

- Experiment, we are all learning
- 2x10 Min Break
- Recording: high demand, few training seats
- Based on interaction
 - Discussions, Exercises + Debriefs
 - Harder in Circuit
 - If I should not call on you, please tell us now



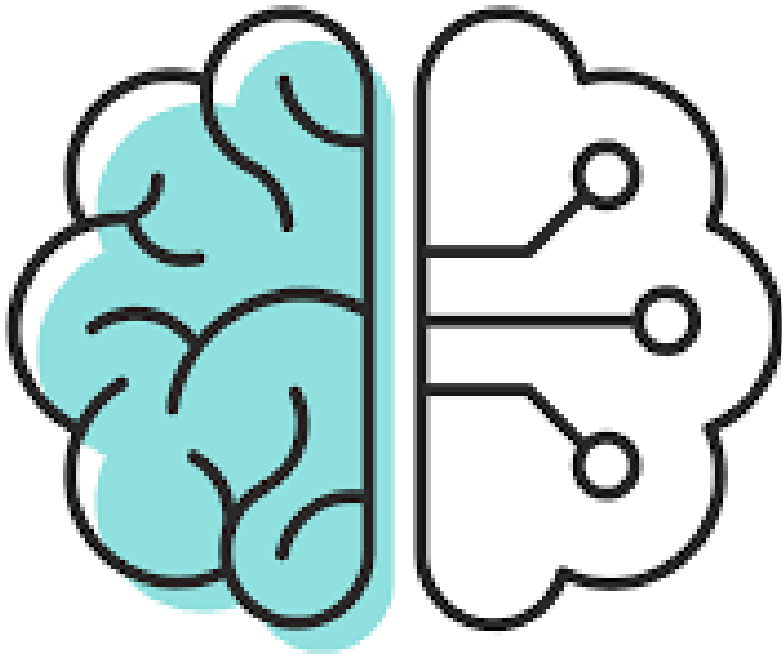
Please Give Us Feedback!



- **Feedback is very important for us. This course is free, but we need to know if this is beneficial for you.**
- **Based on your feedback, we will continue to offer this, develop additional courses – or not.**

Please visit at the end of this course the following website:

<https://forms.office.com/Pages/ResponsePage.aspx?id=PqILJW8f80iQ5uJ2ZmS0d-5LdjYRShZEhN3YmsD7St9UMFZZTFFVQjFSUUxTMzhQNTM3S1hSM0ExVS4u>



- Introduction
- What Is Machine Learning?
- Decision Tree Algorithms
- Limits of Machine Learning
- Neural Networks
- Summary: Definitions and Difference between Important Approaches
- Application of Machine Learning in Our Business

We are Using EnergyAI as a Tool to Run Machine Learning Algorithms



- Please register on EnergyAI before starting the training:(<https://energyai.siemens.cloud/>)
- Request access to “Analyse”
- You will find the files in EnergyAI in the folder shared>users>kay.toepfer@siemens.com>Data Science Workshop.
- More details how to get to Kay’s folder in the following slides



Request Access

Go to Analyse

SIEMENS

Welcome to Energy Analyze!

Fleet Information ☐

Analyse



Run statistical analysis on your data

Data Bucket



Store and share your data files

Data Forum



Discuss data analytics and other technical issues

Documentation



Learn about the applications and services

GraphQL IDE



Explore GraphQL using

Interact



Interact, analyse and

Visualise



Visualise sensor data

Werner

Welcome Kay! I am Werner, your virtual assistant. Please type "help" to see all the available options.

Werner at 09:47:39

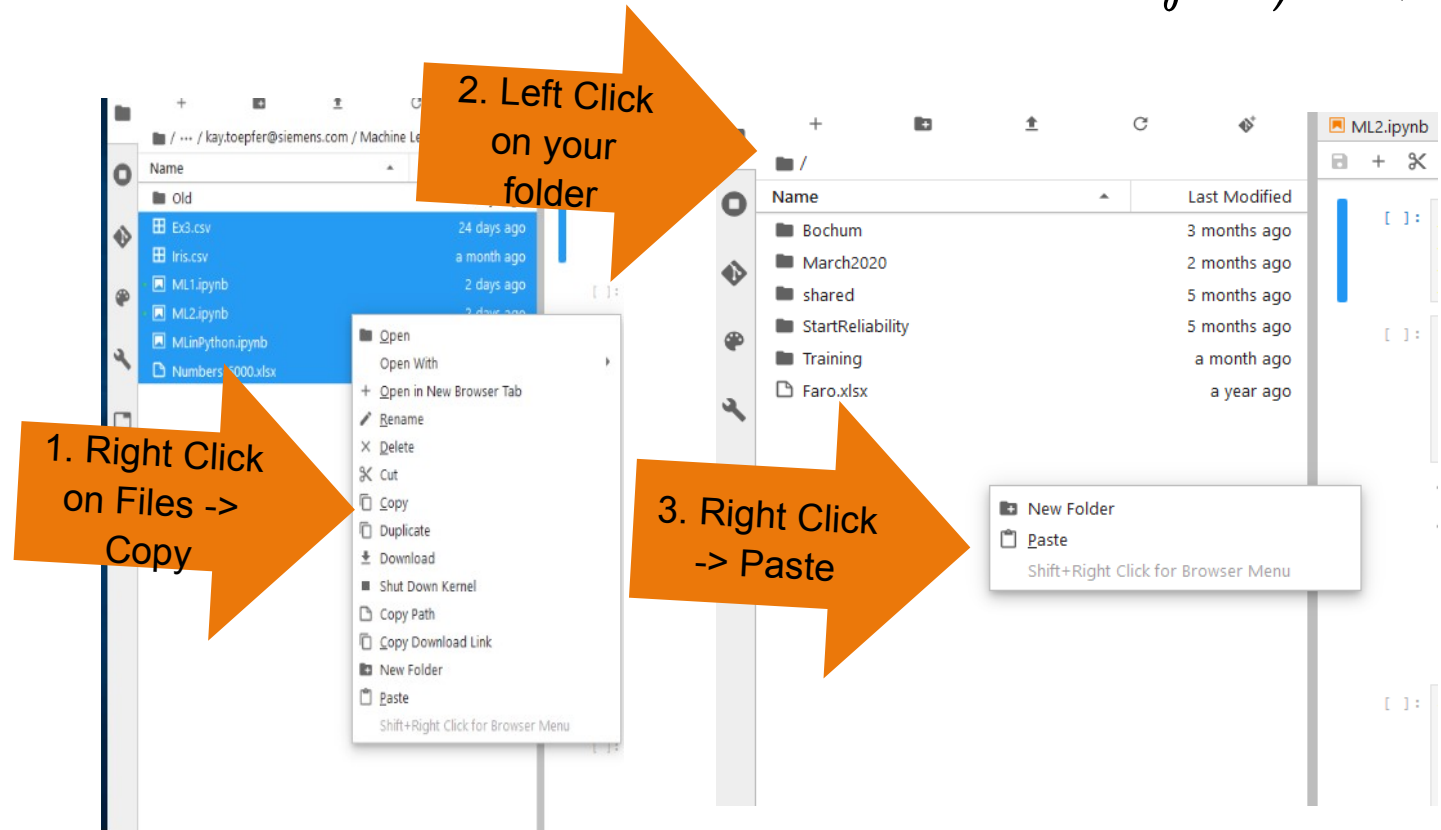
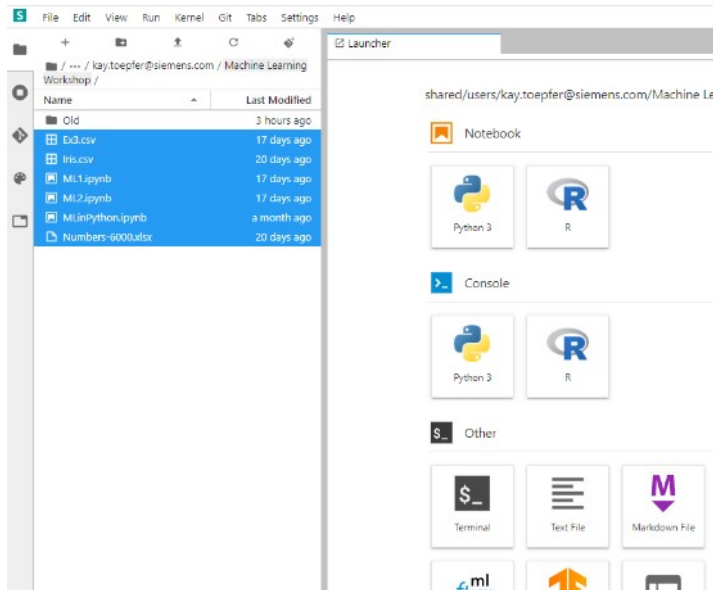
Type your message...

Internal © Siemens AG 2021

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Matthias Groh, SE GP G QPI QM P

Copy All Files in Your Own Folder



Matthias Groh, Your Instructor



- Department “Improvement Projects” of SE Generation Quality
- Data Science consultant
- Six Sigma Master Black Belt

Previous engagements:

- Senior Lean Expert focusing on administration processes
@ Siemens Energy Fossil Products, Berlin, Germany
- Director Six Sigma @ IIL, New York, NY
- Six Sigma Black Belt @ Deutsche Bahn, Frankfurt, Germany

Professional degrees in:

- Precision Mechanics
- Mathematics
- Psychology

Additional qualifications:

- Certified Group Facilitator and Trainer (ABF)
- Certified Six Sigma Master Black Belt (IIL)
- Certified Project Management Professional (PMI)
- Certified Senior Lean Expert (LMI)

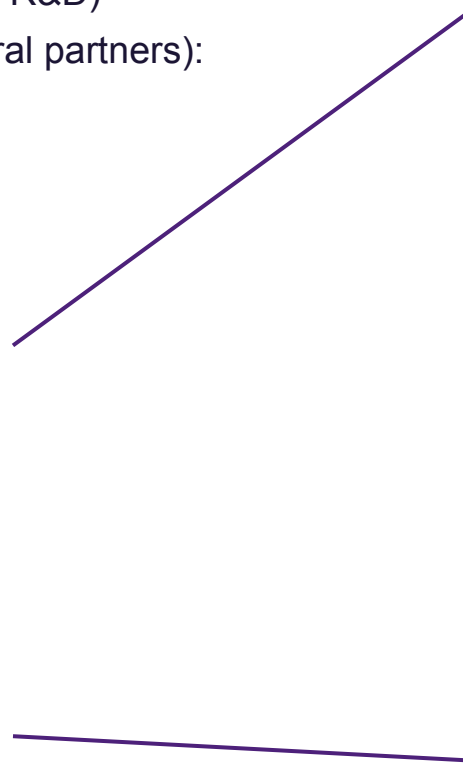


If you need help, please contact our support mail box: dataScienceSupport.energy@siemens.com
To contact me personally, email me at groh.matthias@siemens-energy.com

Introduction: Data Science @ SE GP G QPI QM P Improvement Projects

We help your employees getting the skills for the digital future:

- Global Data Science Skill Network (with Matt Bryan, LGT R&D)
- Data Science and Machine Learning trainings (with several partners):
 - Currently 11 trainings developed and executed
 - Attended by >500 participants mostly from R&D
- Analytic services in specific improvement projects
- EAI CodeShare (with Thomas Buller, LGN R&D)



Data Science @ QPI QM P:



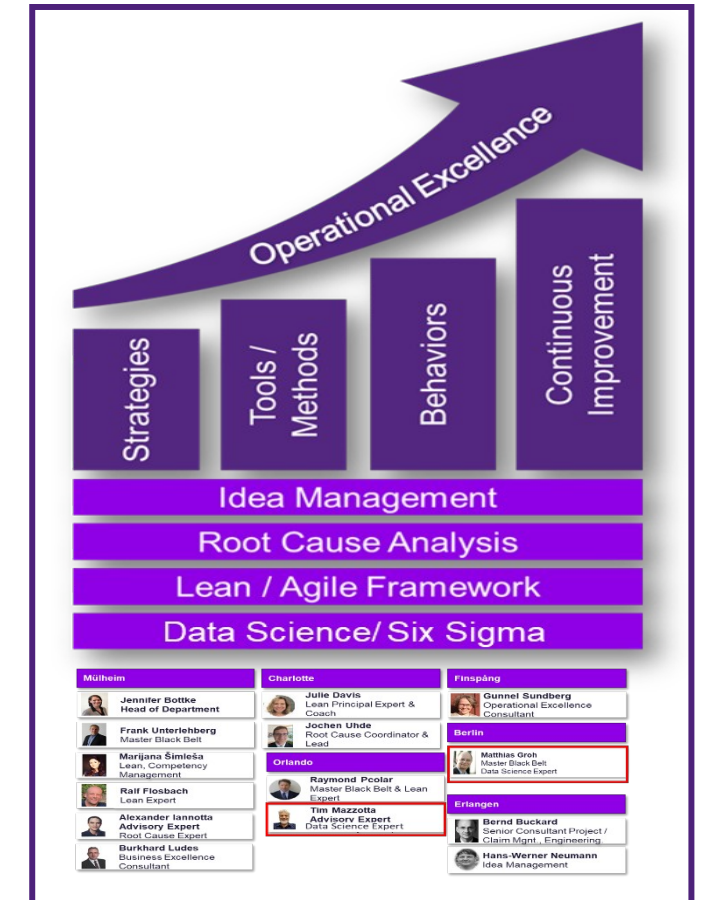
Matthias Groh
Master Black Belt
Data Science Expert



Tim Mazzotta
Advisory Expert
Data Science Expert

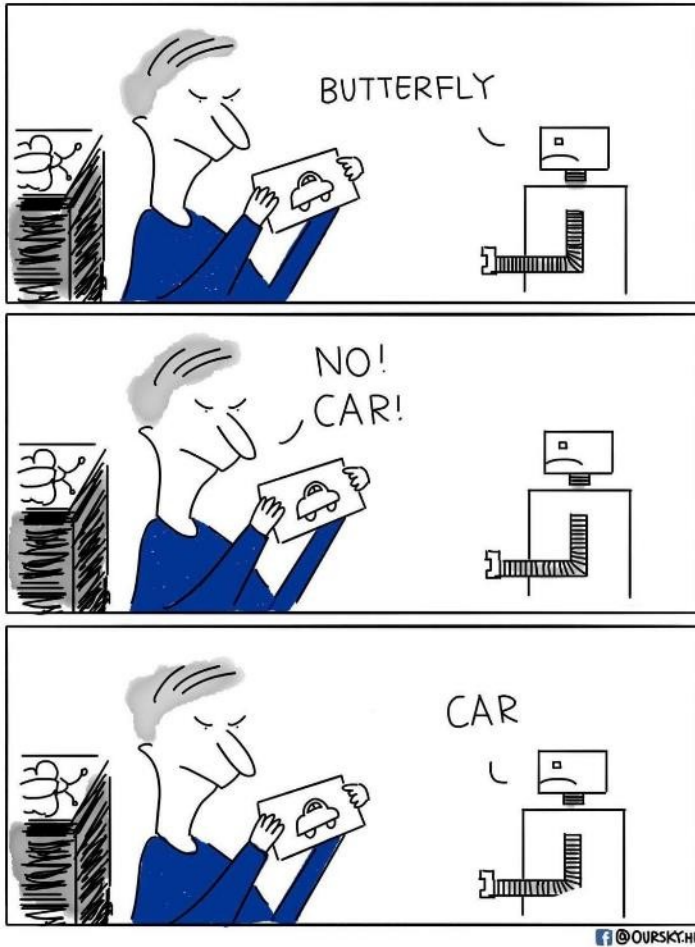
Nicolas Schneider
Coop Master Student

Kay Toepfer
Master Student

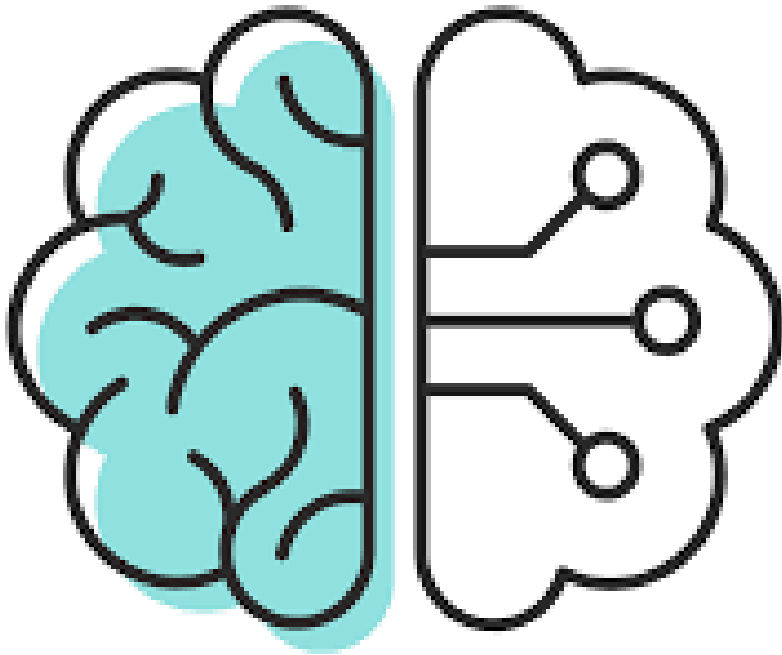


We help building the bridge between Data Science and daily engineering business.

Please Quickly Introduce Yourself



- Your name and function / organization
- Your background regarding Machine Learning
- Which site



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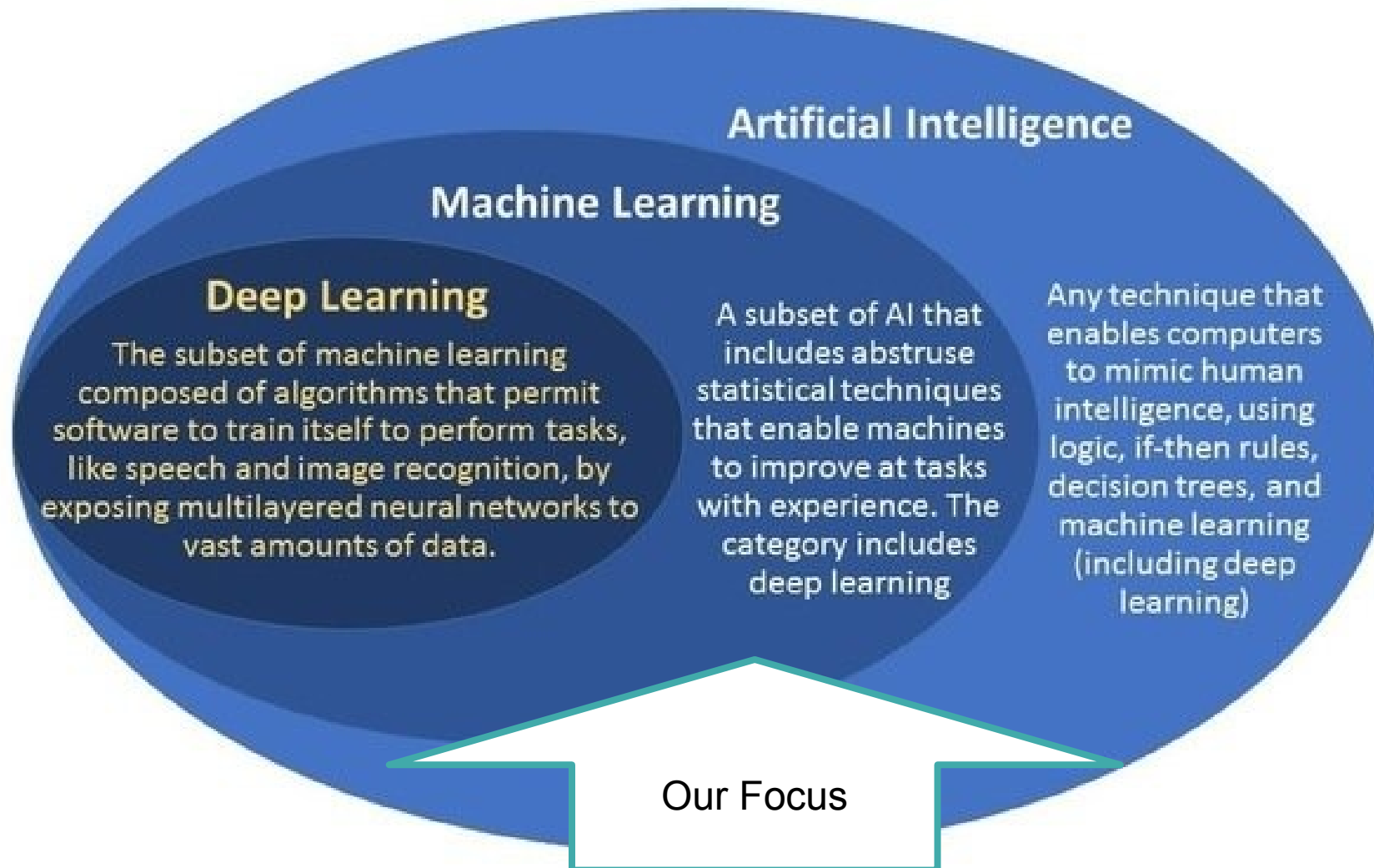
What is Machine Learning?

The First Sentences from Wikipedia ...

Machine learning (ML) is the scientific study of **algorithms and statistical models** that computer systems use to effectively perform **a specific task without using explicit instructions**, relying on patterns and inference instead. It is seen as a subset of artificial intelligence. Machine learning algorithms **build a mathematical model based on sample data, known as "training data"**, in order to **make predictions or decisions** without being explicitly programmed to perform the task.

https://en.wikipedia.org/wiki/Machine_learning

A Trial to Distinguish a bit between Some Key Fields



In this Workshop We Will Use a Tool Based on Python and Scikit-learn

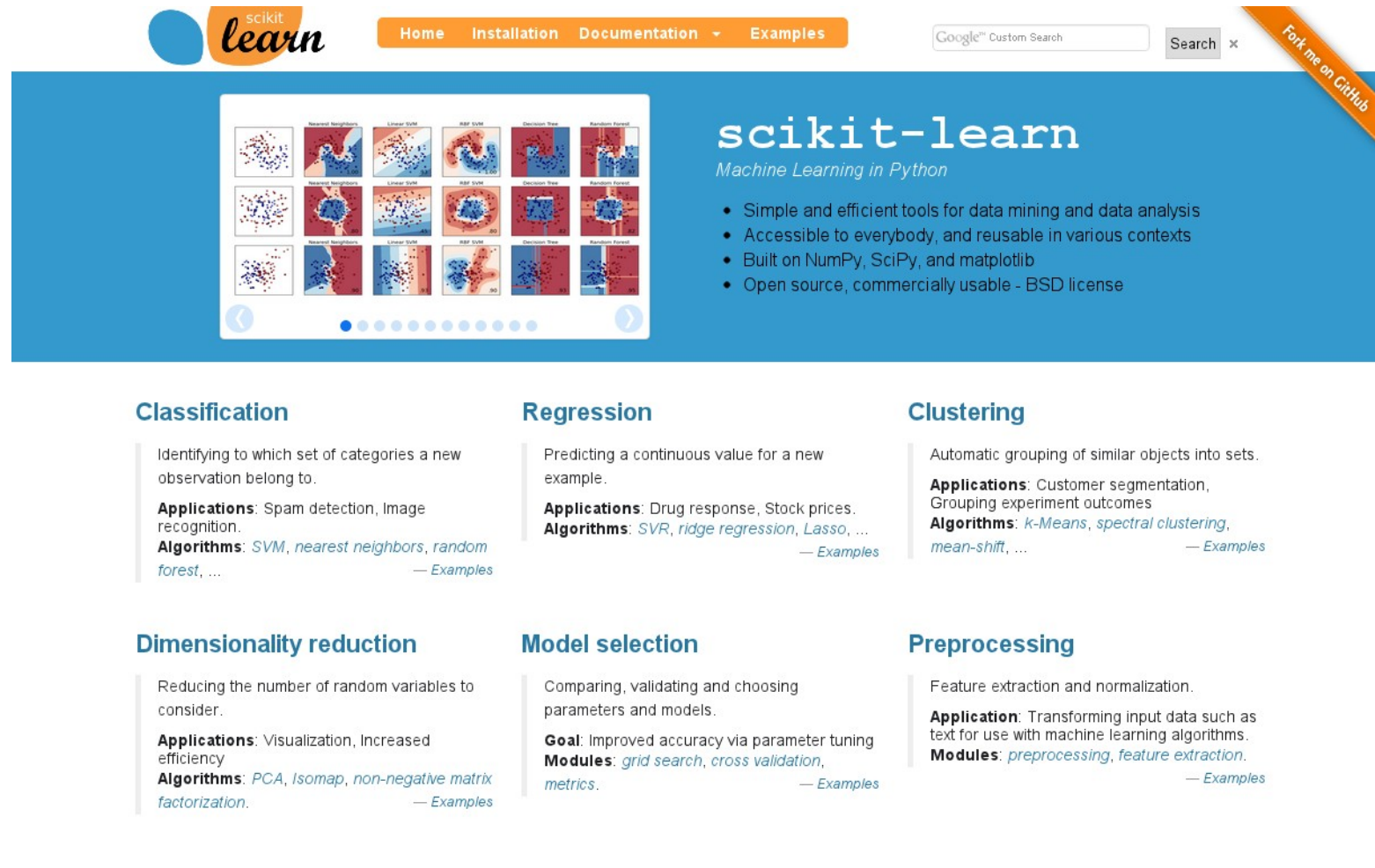


Python is an interpreted, high-level, general-purpose programming language, leading in data science.



Scikit-learn is a free and powerful software machine learning library for Python. It is featuring various classification, regression and clustering algorithms

On www.scikit-learn.org There Is Abundant Information on the Complete Library



scikit-learn
Machine Learning in Python

- Simple and efficient tools for data mining and data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying to which set of categories a new observation belong to.

Applications: Spam detection, Image recognition.

Algorithms: SVM, nearest neighbors, random forest, ... — Examples

Regression

Predicting a continuous value for a new example.

Applications: Drug response, Stock prices.

Algorithms: SVR, ridge regression, Lasso, ... — Examples

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, Grouping experiment outcomes

Algorithms: k-Means, spectral clustering, mean-shift, ... — Examples

Dimensionality reduction

Reducing the number of random variables to consider.

Applications: Visualization, Increased efficiency

Algorithms: PCA, Isomap, non-negative matrix factorization. — Examples

Model selection

Comparing, validating and choosing parameters and models.

Goal: Improved accuracy via parameter tuning

Modules: grid search, cross validation, metrics. — Examples

Preprocessing

Feature extraction and normalization.

Application: Transforming input data such as text for use with machine learning algorithms.

Modules: preprocessing, feature extraction. — Examples

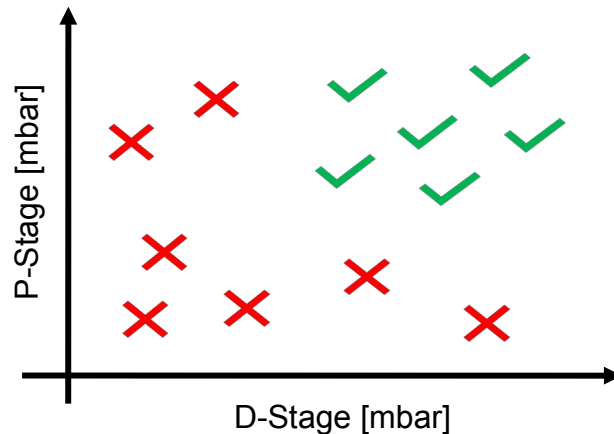
2 – Learning Types

Supervised Learning

“Features” (Xs) and “Targets” (Ys)

Example: Start Reliability

Y: Successful ignition ✓ / Failure ignition ✗



Result: Model to predict Y: $Y = f(X_1, \dots, X_n)$
i.e. for assuring next start is reliable

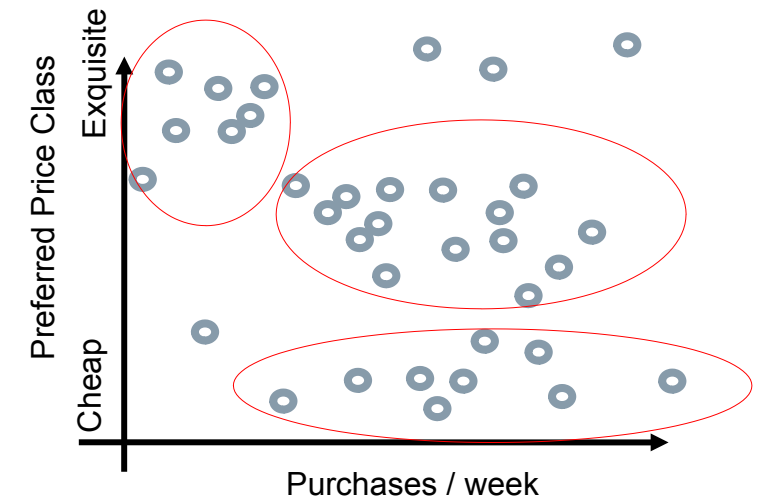
X and Y are known

Only X are known

Unsupervised Learning

Only “Features” (Xs)

Example: Amazon Customer analysis



Result: Clusters to group customers in,
i.e. for targeting advertisement

For our business „supervised learning“ is probably the most frequent relevant learning type

Challenges of Machine Learning Application in Our Business

Consumer Data Analysis

- Little knowledge about consumer.
- Users can be tracked very closely.



Turbine Data Analysis

- Starting from deep understanding -> very difficult to add value on top.
- Limited number of engines + change in time, operating conditions, ambient conditions.
- ML tools work best when “output data” exists (“supervised” vs. “unsupervised” learning).



We still have to learn together in which situations we can extend our knowledge with ML.

ML Application Example: Compressor Icing Monitoring

Lead: Stephan Klumpp



Initial situation & Problem

- During GT Tests installed cameras “see” how much ice occurs on compressor blades
- Someone has to watch ours of video

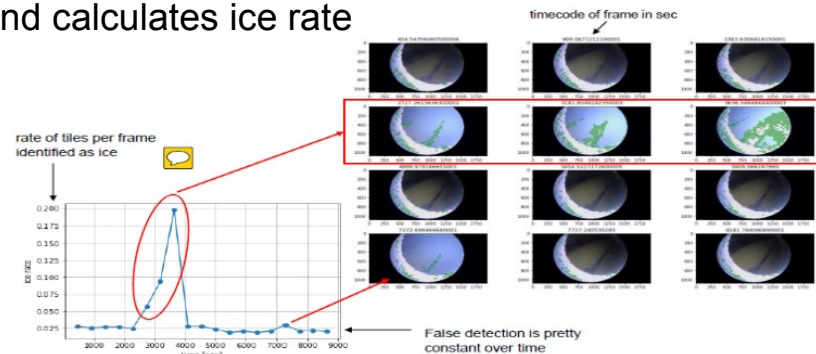


Approach

- Break up video pictures into tiles: Algorithm analyzes each tile, not entire picture (120 pictures with 42.000 tiles)
- Break up tiles into 30x30 pixels
- Label a sample manually: do they have ice or not
- Build Model with Convolutional Neuro Network

Solution

- NN Model analyzes the video data and calculates ice rate



Benefits

Benefits Machine Learning over trad. stat. models

- Enable to analyze 900 pixels per tile
- Enabled to analyze 42.000 tiles
- High accuracy of 90%

Benefits Business

- 2 Hz picture rate can be analyzed by standard computer in real time
- Eliminates engineering hours + frustration
- More test hours can be measured -> more data

For more information: contact stephan.klumpp@siemens.com

ML Application Example: GT AutoTuner - Emission Module

Owners: Katja Rietbrock (Commercial) Sterzing Volkmar (Technology), Gurdev Singh (Technical)



Initial situation & Problem

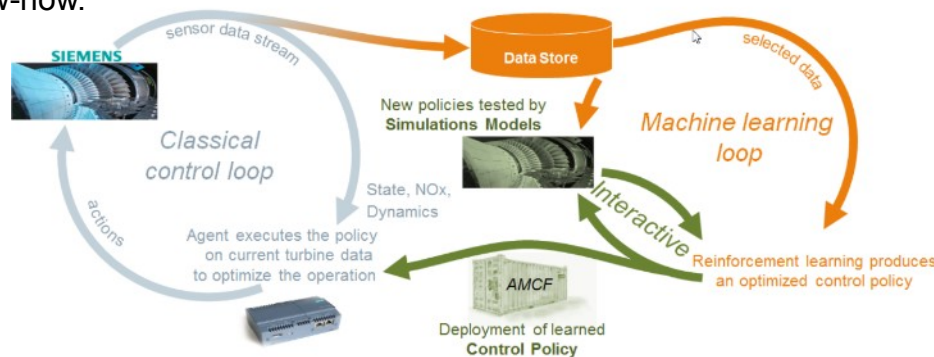
- GT tuning:
 - setting amount + gradient of different gas stages in the burner during operation from 70% - baseload
 - Many input parameters incl. gas quality, ambient weather conditions etc.
 - Many output variables, incl. emissions (NOx, CO) Vibration of parts, power
- Complexity comes from:
 - Complex fluid dynamics and chemical processes during combustion
 - Difficult to model during design
 - Change in ambient conditions
 - Degradation of parts and hardware upgrades
 - Sensor noise
 - Optimization has to be done within short time/ real-time
- Tuning needs lots of expert resources without necessarily finding the best optimum.

Approach

- Studied historical data from the GT fleet
- Worked with several algorithms incl. Recurrent NN and Reinforcement Learning
- Use of signal processing, numerical as well as T3000 simulator, visualizations, and domain experience
- Over 10 years

Solution

- Provide updates to the traditional T3000 control loop by “Reinforcement Learning” results fused with domain know-how.
- Installed in 15 GTs



Benefits

- Better year round control of emissions and dynamics with an average reduction of NOx ~ 10 ppm or higher.
- AS3000 platform used for GTAT can be used to host other ML based uses cases in future.

Having a „Supervised Learning“ Approach, We Should Distinguish between the Different Data Types of the Target Variable

The target variable is

Attributive

Numerical



success



failure

40,45 39,97
40,85 38,93
39,12 40,07

The learning approach is called:

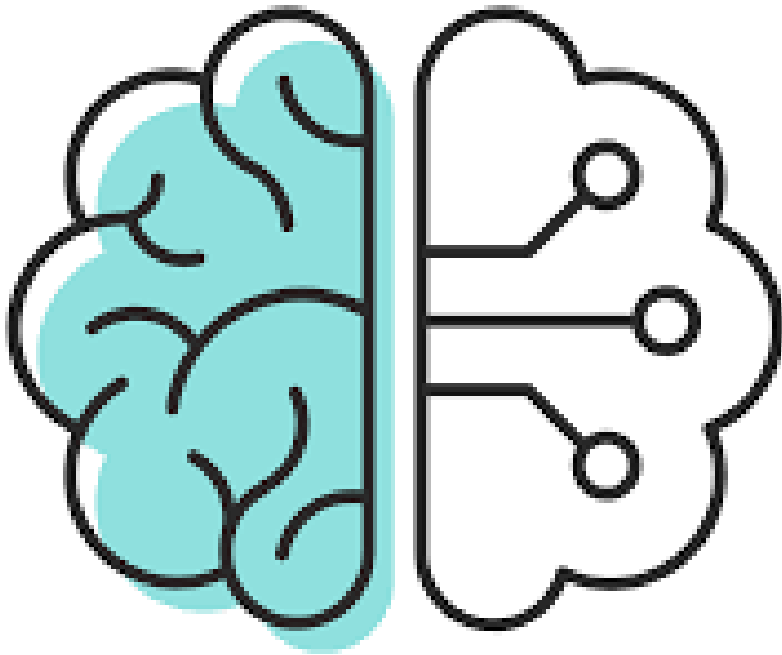
„Classification“:

Selecting the right group for a new element

„Regression“: Estimates a numerical value for a new element

**Most learning algorithms have a version for each,
Classification and Regression**

**We focus on Classification examples, because they are easier to explain.
Regression follows the same principals.**

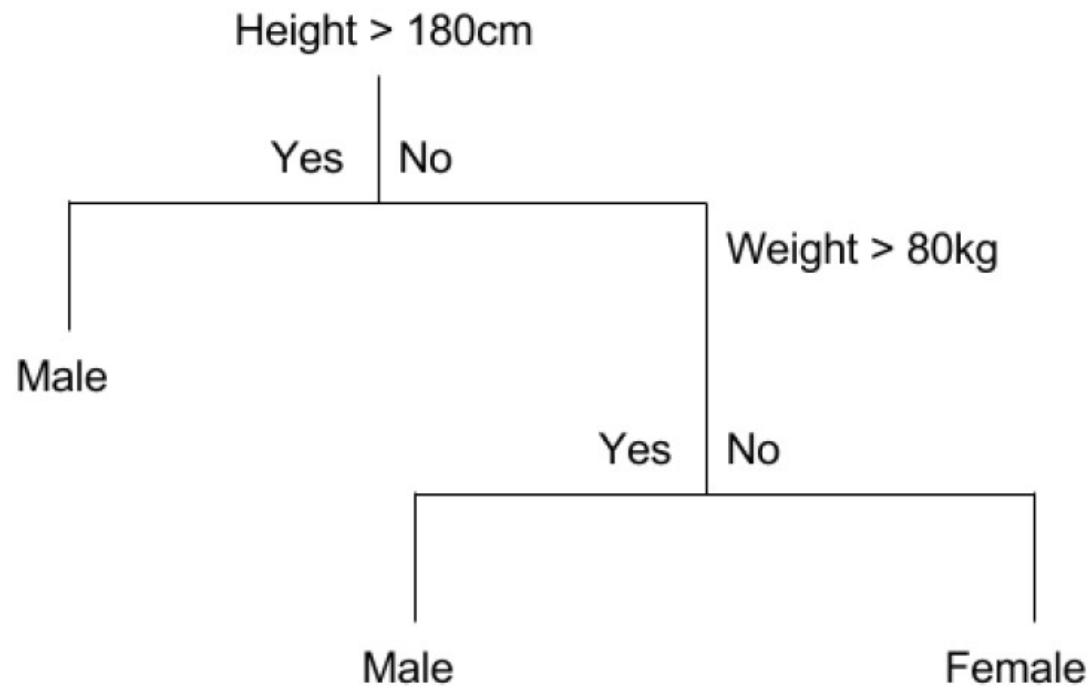


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Classification by Decision Trees

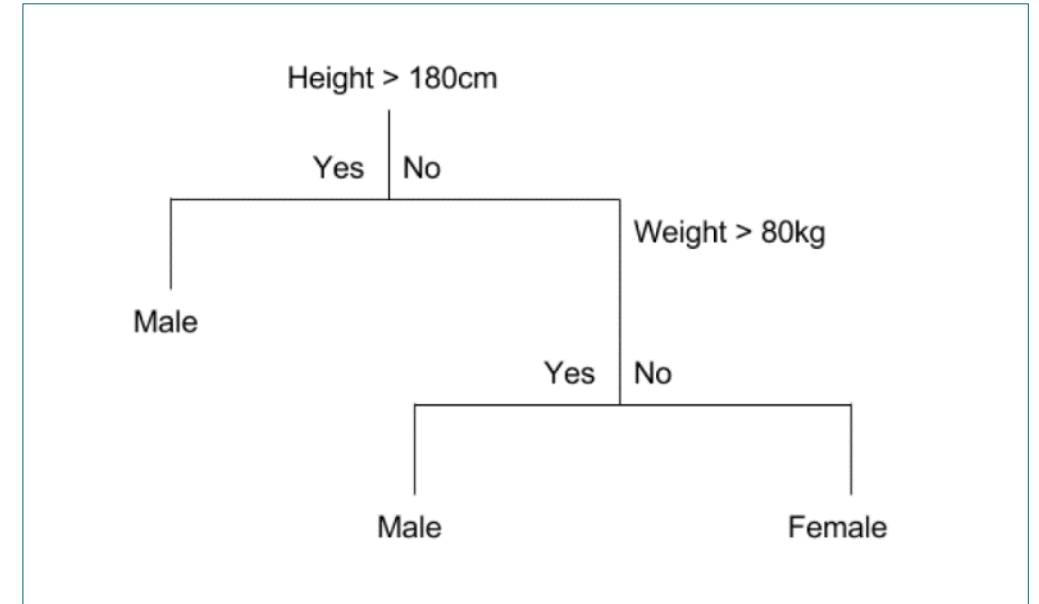
New wording CART (Classification and Regression Tree)

A crude example:



Main Idea How a Decision Tree Is Build

1. Start at the top: first node
2. Evaluate:
 - all input variables
 - all their possible split points
 - Which would have the highest purity
3. Split accordingly and generate two leaves
4. Repeat 2 + 3 with both leaves
5. Stop when:
 - Purity can't be further improved
 - Some predefined metrics are reached:
 - Maximal depth (=consecutive branches) ☾ will play with this
 - Maximal # of leaves

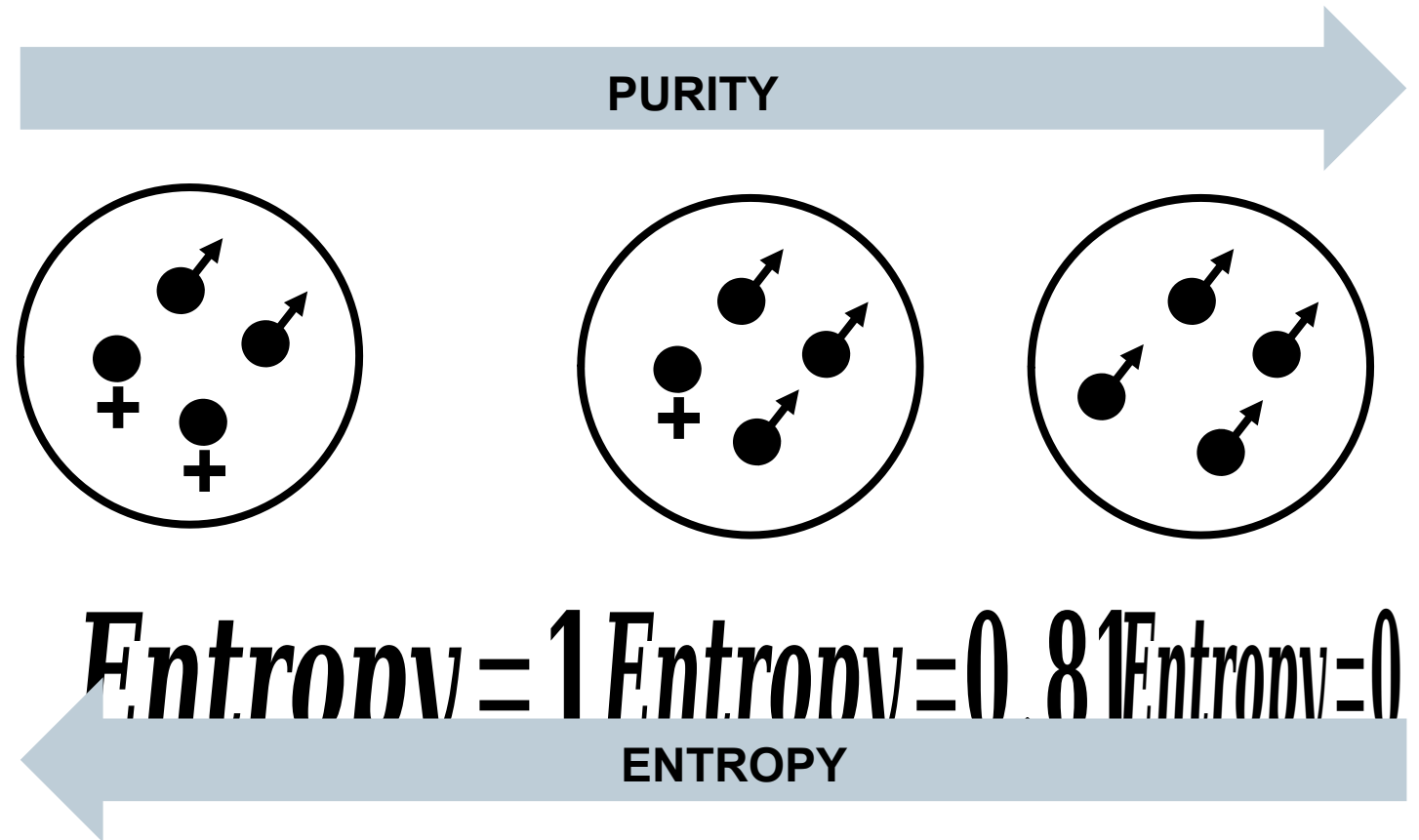


Purity of a Dataset

Entropy is used to measure purity:

- Measures actual Impurity
- Entropy = 0: perfectly pure node(s)
- Entropy ≥ 1 : large mix of males and females

Goes back to the Shannon Entropy in physics



With Decreasing Entropy the Purity Increases.

Iris Data Set: 75 Data Points about 3 Types of Irises

21x Iris Setosa



27x Iris Versicolor



27x Iris Virginica



Measured length and width of leaves:

Sepal (dt: Kelchblatt)

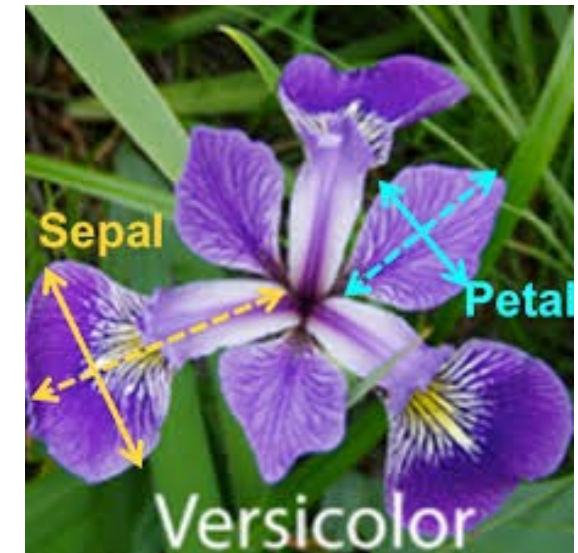
Petal (dt. Blüten/ Kronblatt)

X[0]: SepalLengthCm

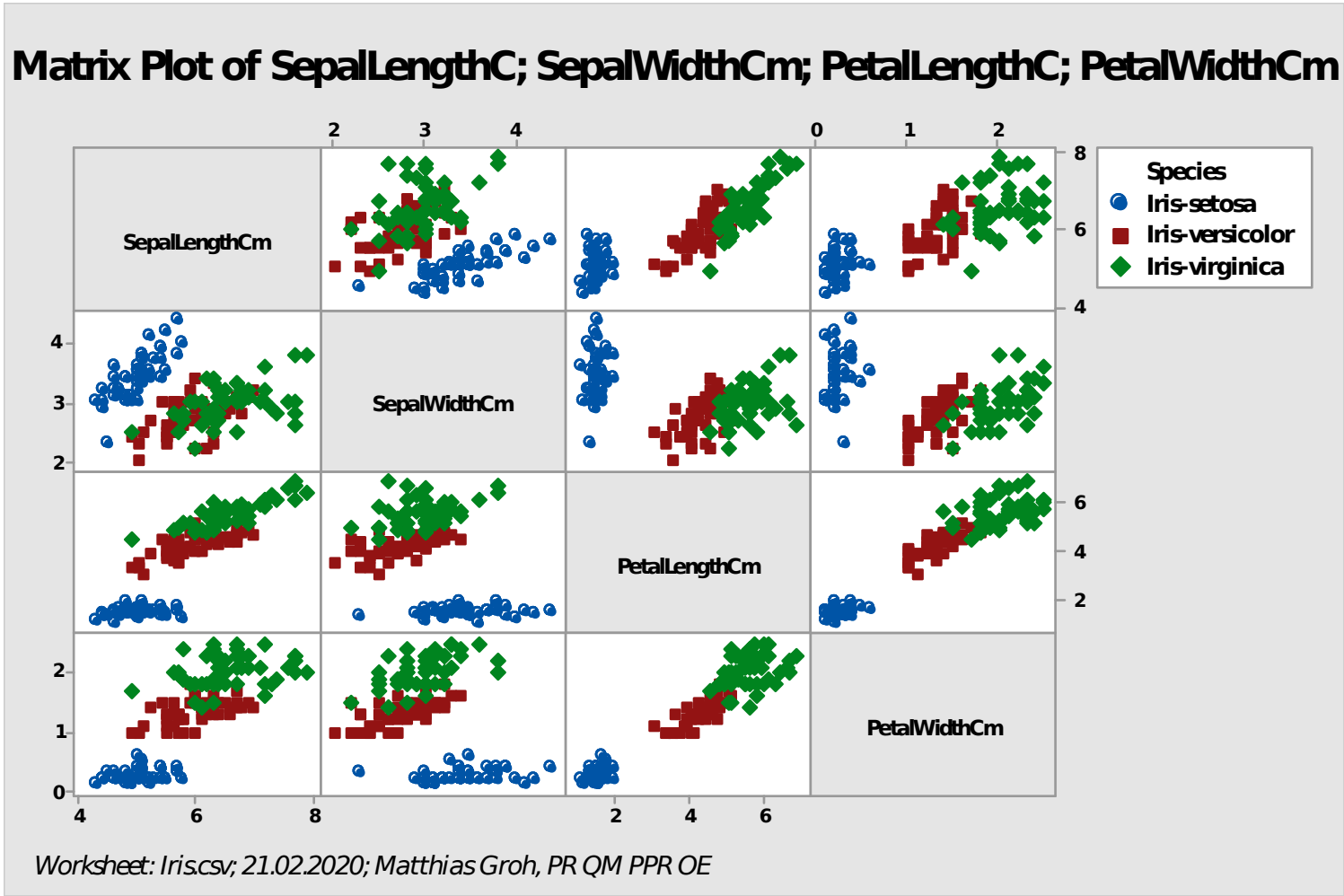
X[1]: SepalWidthCm

X[2]: PetalLengthCm

X[3]: PetalWidthCm



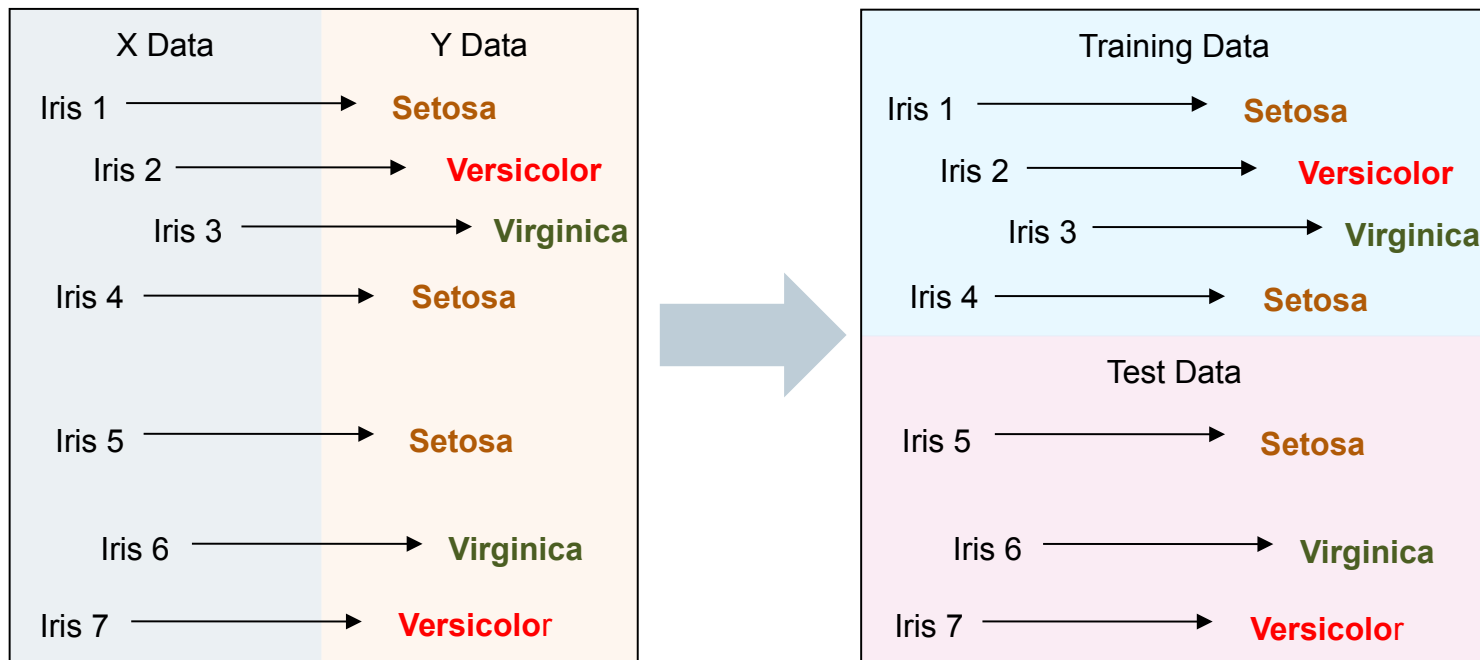
Plot Your Data!



Training Data and Test Data

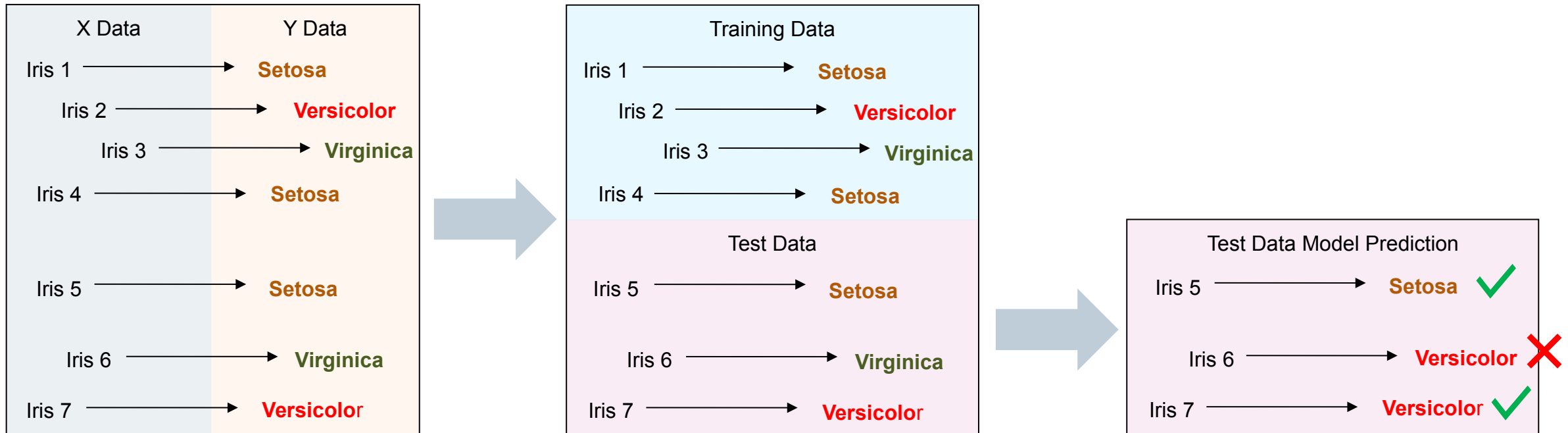
Split your data in training and test data:

- Training data: to train the model
- Test data: to validate if the model works on data the model has not seen yet



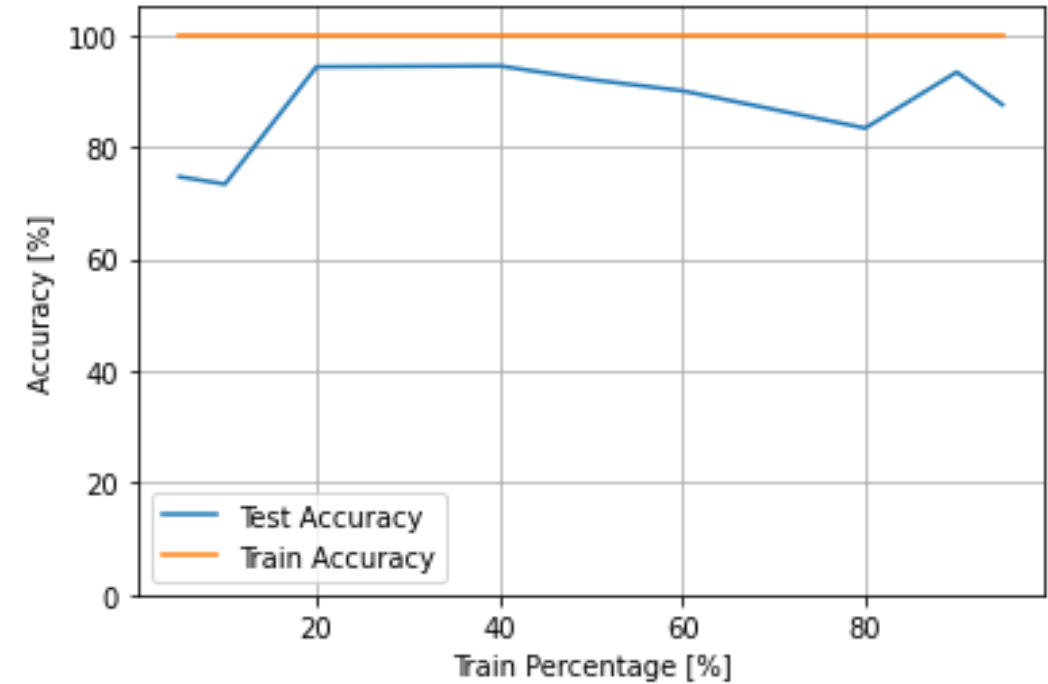
Accuracy of Test Data

- Accuracy measures how correctly your model predicts the test data
- In our example



Accuracy of Train Data

- Sometimes it's useful to see how good the model is on the test and on the train data



How to Jupyter Notebook: Editing Tool for Python

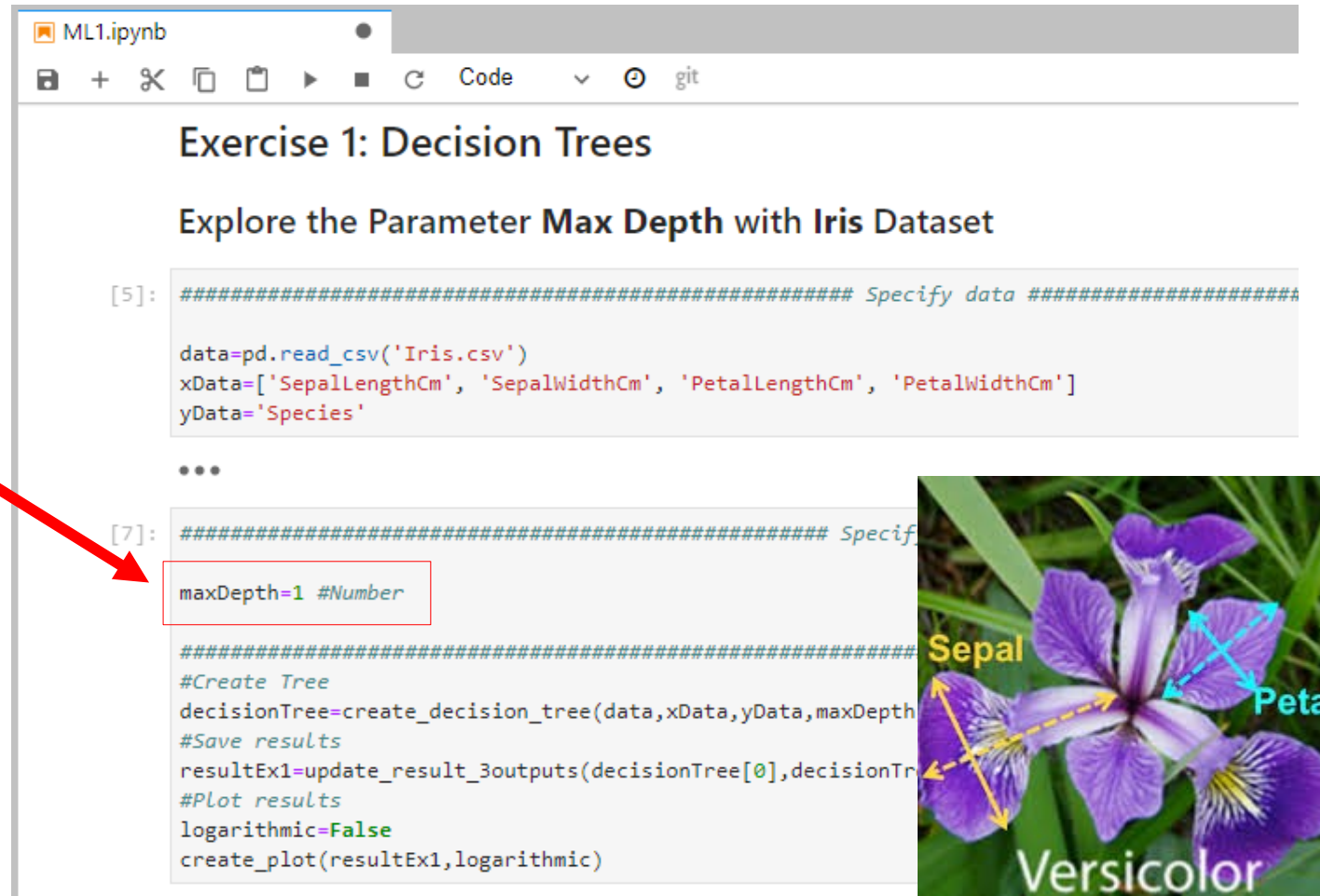
- Every box is program code which you can run with shift + enter
- There's a symbol in front of each box
 - [] means the cell is not executed
 - [*] means the cell is executed and the code is running
 - [3] means that the calculation is completed (continuous numerating of executed boxes)
- After running the exercise cells, your results will be saved
- If you've made a mistake, run the entire notebook again (at least the first 4 lines, then every input / result will be reset)

Exercise 1: Explore the Parameter Max Depth

1. Open the ML1.ipynb Notebook
2. Run through all cells till Exercise 1: Decision Trees (shift+enter)
3. Change MaxDepth to 1
4. Run the cell

Repeat:

- with different Max Depth: 1, 2, 3, 4, ... 10
Run the cell every time with shift+enter.
Don't change anything else so that we can compare.
 - Consider the change in the accuracy of training and test data sets and the entropy for different maximum depths.
5. Compare accuracy of training and test data sets.
Which difference do you see?
 6. Run the next two cells (till `show_trees(resultEx1)`)
and have a look at the trees. What development of
entropy, accuracy do you see?



```
[5]: ##### Specify data #####

data=pd.read_csv('Iris.csv')
xData=['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']
yData='Species'

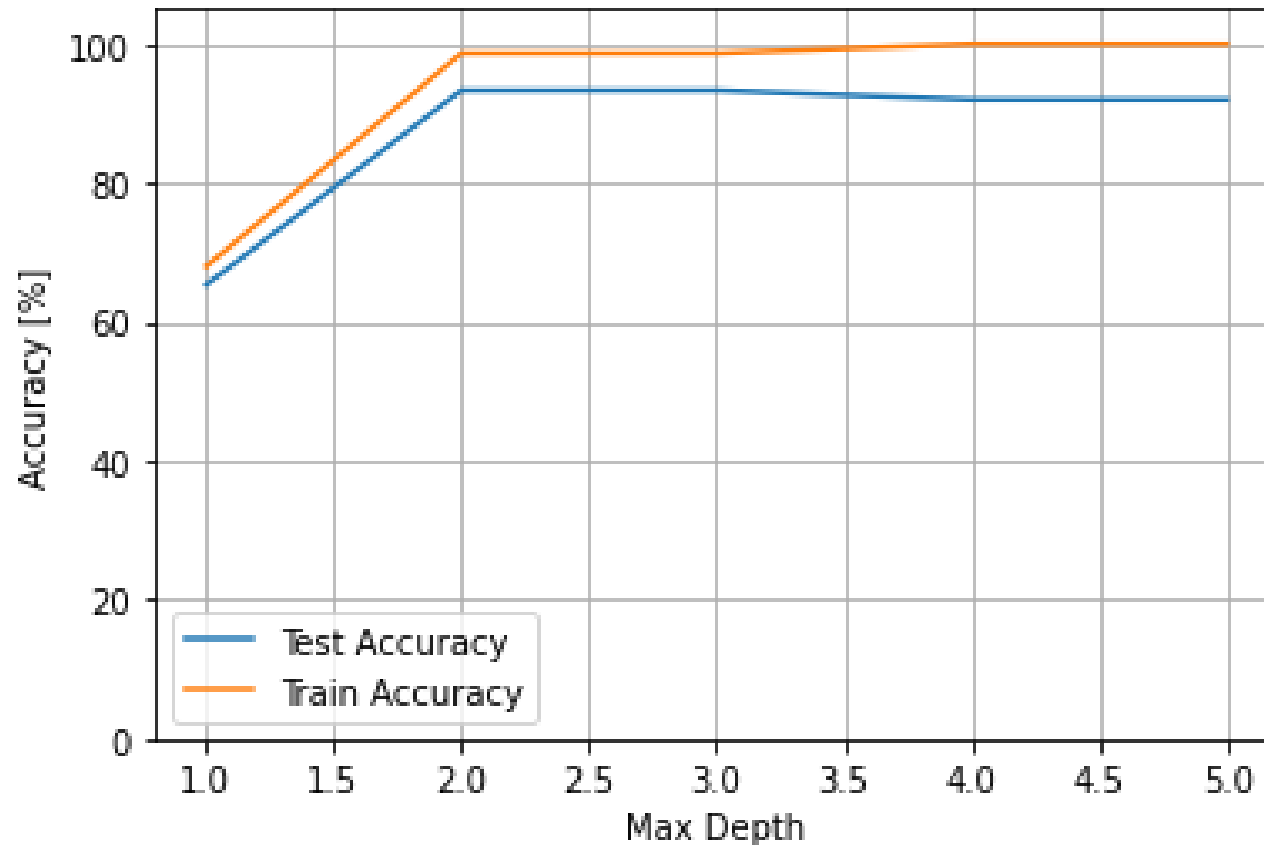
...

[7]: ##### Specify

maxDepth=1 #Number

#####
#Create Tree
decisionTree=create_decision_tree(data,xData,yData,maxDepth)
#Save results
resultEx1=update_result_3outputs(decisionTree[0],decisionTr
#Plot results
logarithmic=False
create_plot(resultEx1,logarithmic)
```

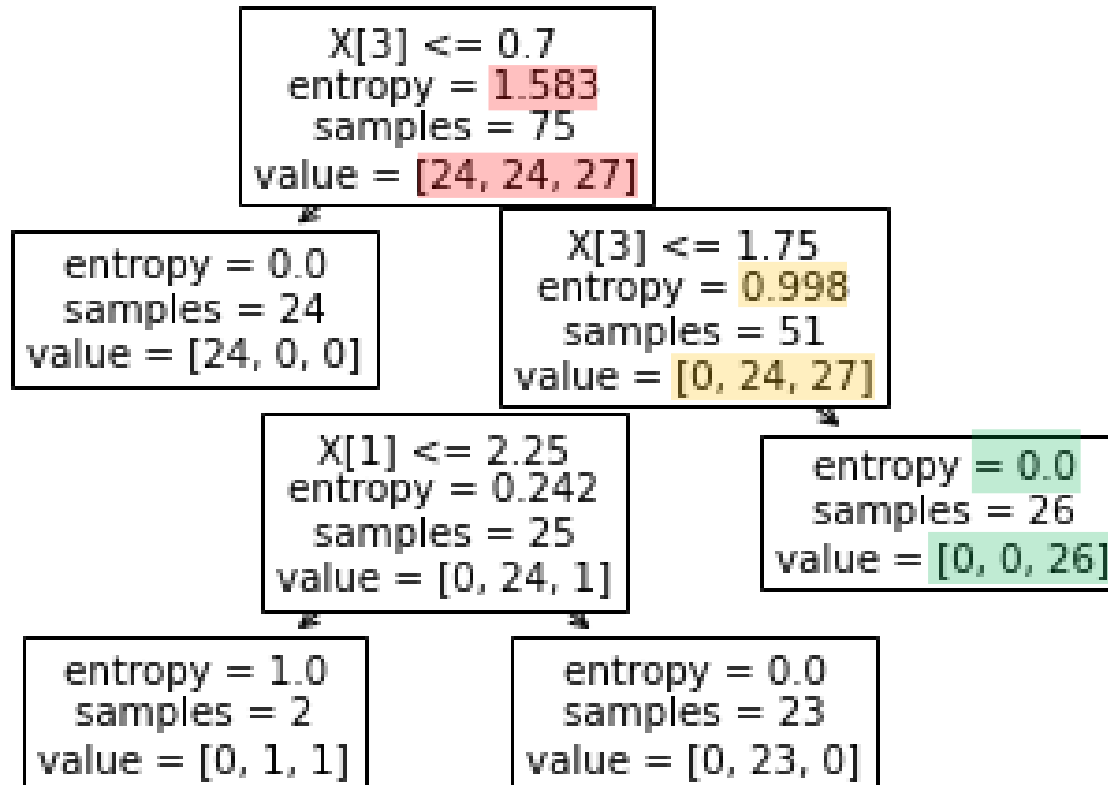
Explore the Parameter Max Depth



- At the beginning the accuracy increases with increasing max depth
- For max depth > 2 the accuracy don't change very much

Explore the Parameter Max Depth

Max Depth: 3



Entropy decreases for “purer” nodes

Example 2: Number Recognition

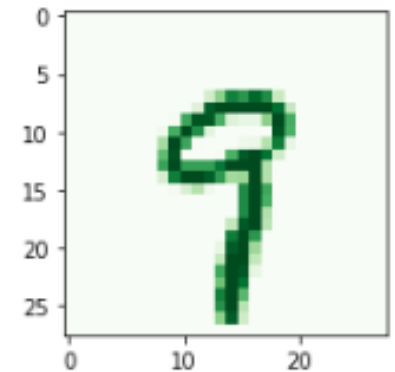
X-Variables: $28 \times 28 = 784$ variables with pixel data

X-Range: 0-255 for intensity of the pixel

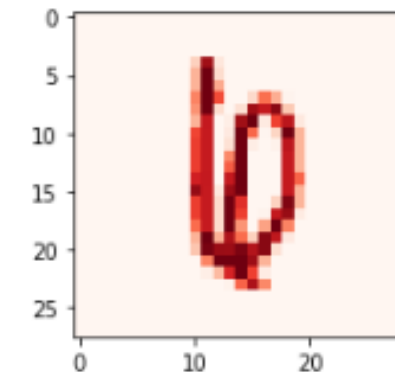
Y-Variable: correct number: 0, 1, ..., 9

Numbers (700): 630 training data + 70 test data

Easy to identify:



Hard to identify:



Exercise 2: Numbers Recognition

Find the Optimal Tree Depth

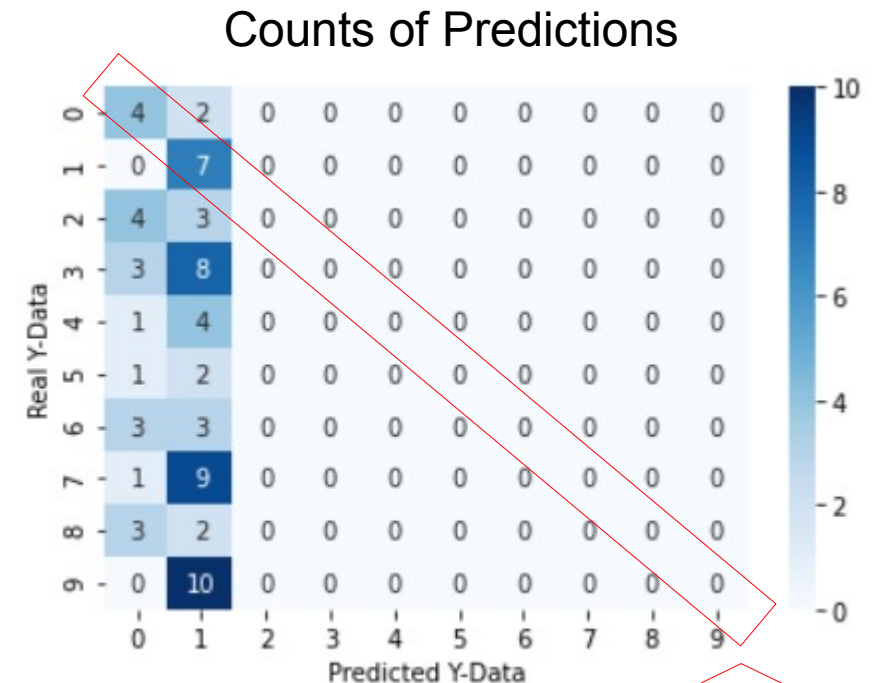
1. Go to Exercise 2: Decision Trees
2. Perform decision tree with different Max Depth (1 – 20) and find the optimal depth
 - What else the optimal depth?
 - What else do you observe?

Exercise 2: Decision Trees

Explore the Parameter **Max Depth** with **Numbers Recognition Dataset**

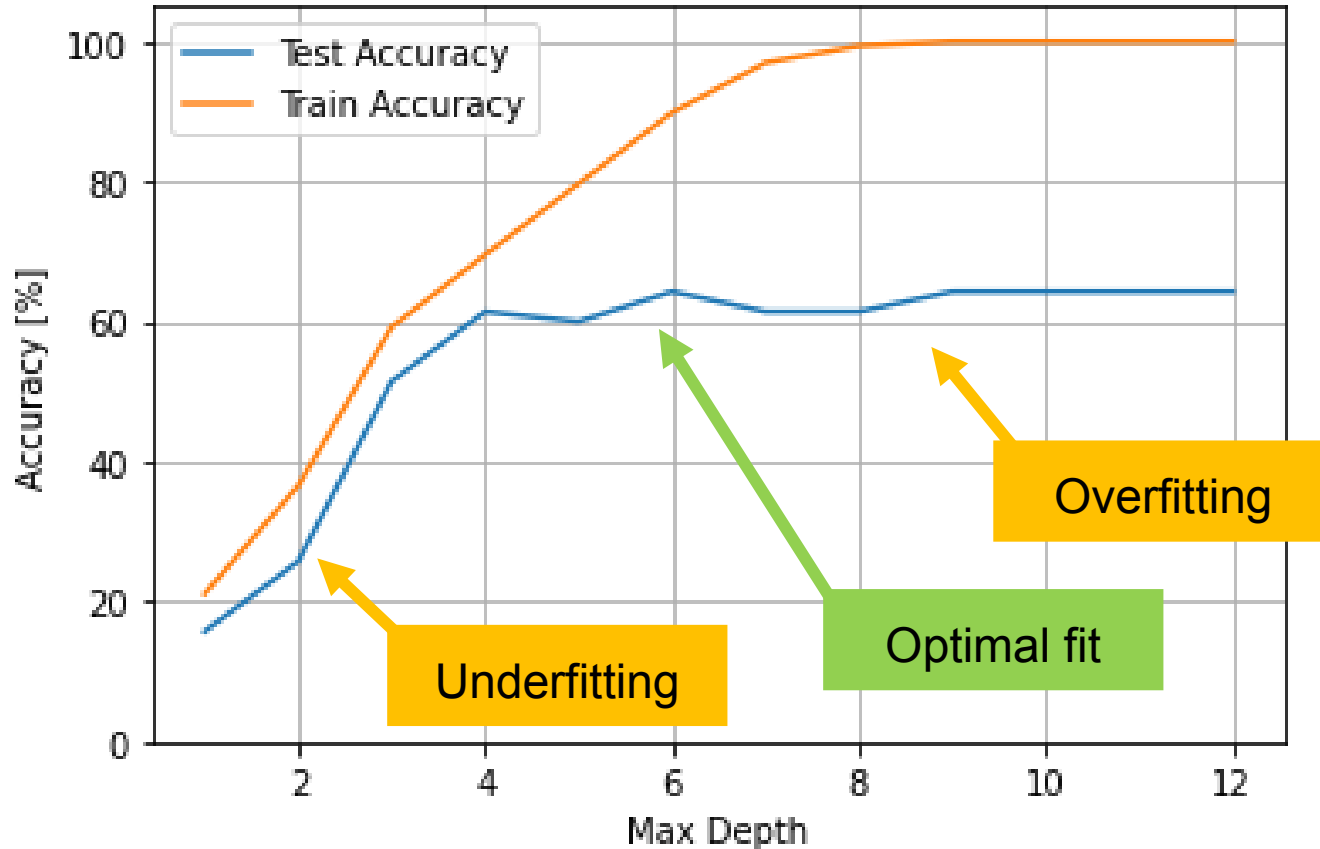
```
##### Specify data #####
data=pd.read_excel('Numbers-700.xlsx')
xData=[str(i) for i in list(range(1,85))] #Generates a list ['1', '2', '3', ..., '783', '784'], these are the column names
yData='Number'
...
##### Specify parameters #####
maxDepth=5 #Number
#####
#Create Tree
decisionTree=create_decision_tree(data,xData,yData,maxDepth,trainPercShare,randomShuffle,randomState)
```

This head chart shows you what happens with the test data when you apply the model:



Diagonal =
correct
predictions

Exercise 2: Number Recognition Using Decision Tree



Optimal max depth around 6

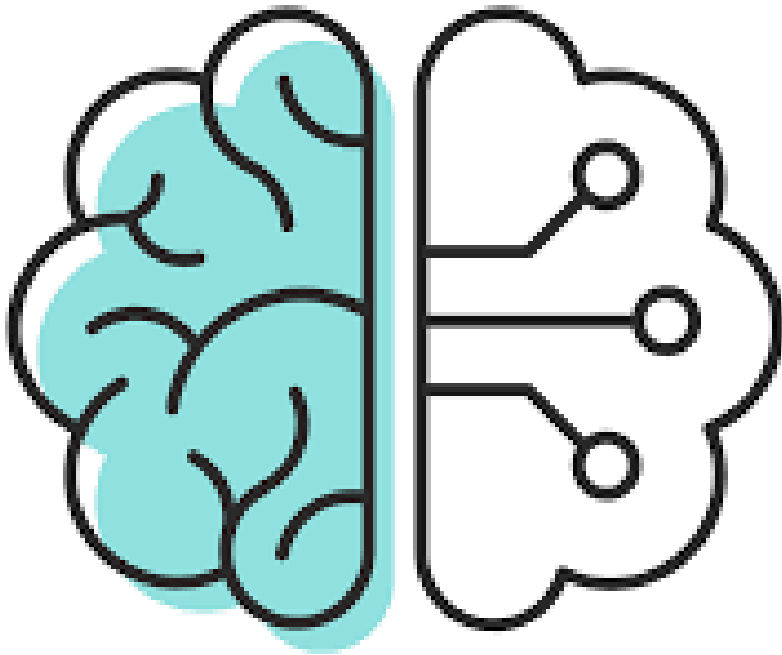
Overfitting

Optimal fit

Underfitting

Over- and underfitting will be explained in the next section.

For more information go to: <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html#sklearn.tree.DecisionTreeClassifier>



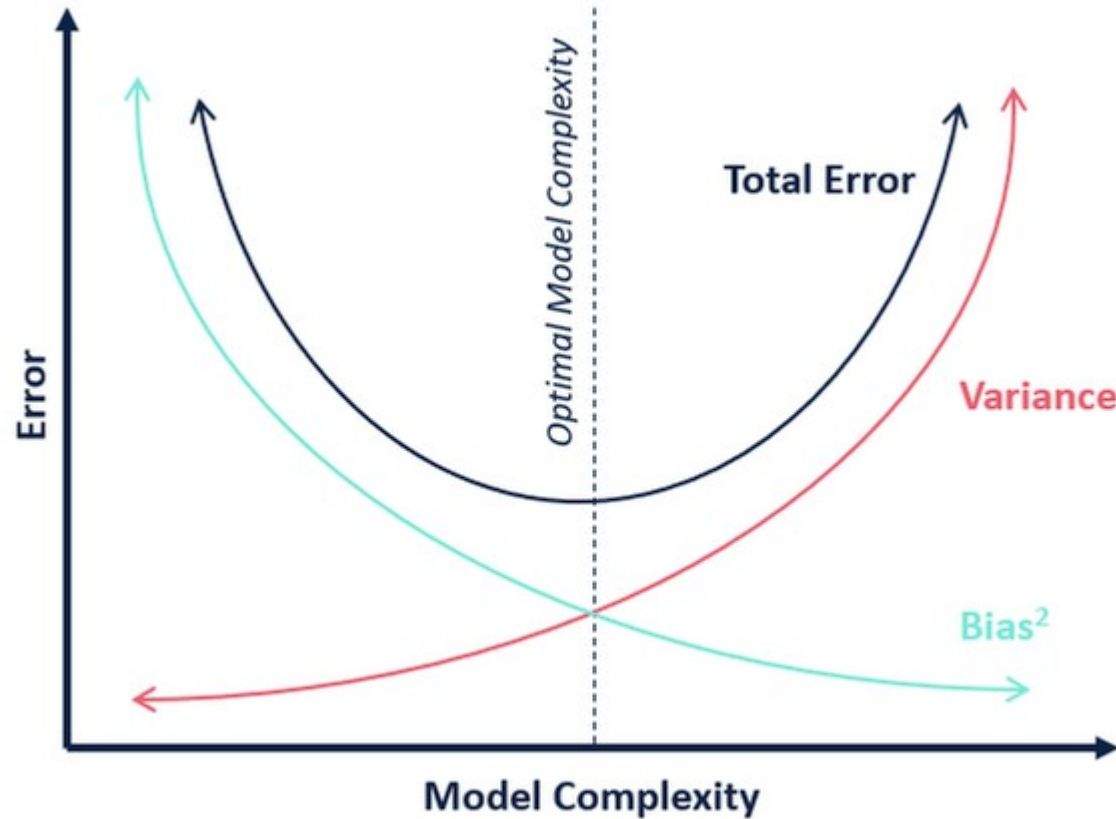
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Model Limitations

	Error Type	Summary	Details	How to prevent it
Model Error	Bias Error/ Underfitting	Model is too simple	The (unknown) true target function is more complex than the model.	• Review with domain experts • More complex analysis
	Variance Error/ Overfitting	Modell is too complex	The true target function is simpler than the model. Common reason: model tries to explain randomness	• Review with domain experts • Split data in training and test data • Talk to data scientist

Any build model build is an approximation of a true target function and therefore includes errors.

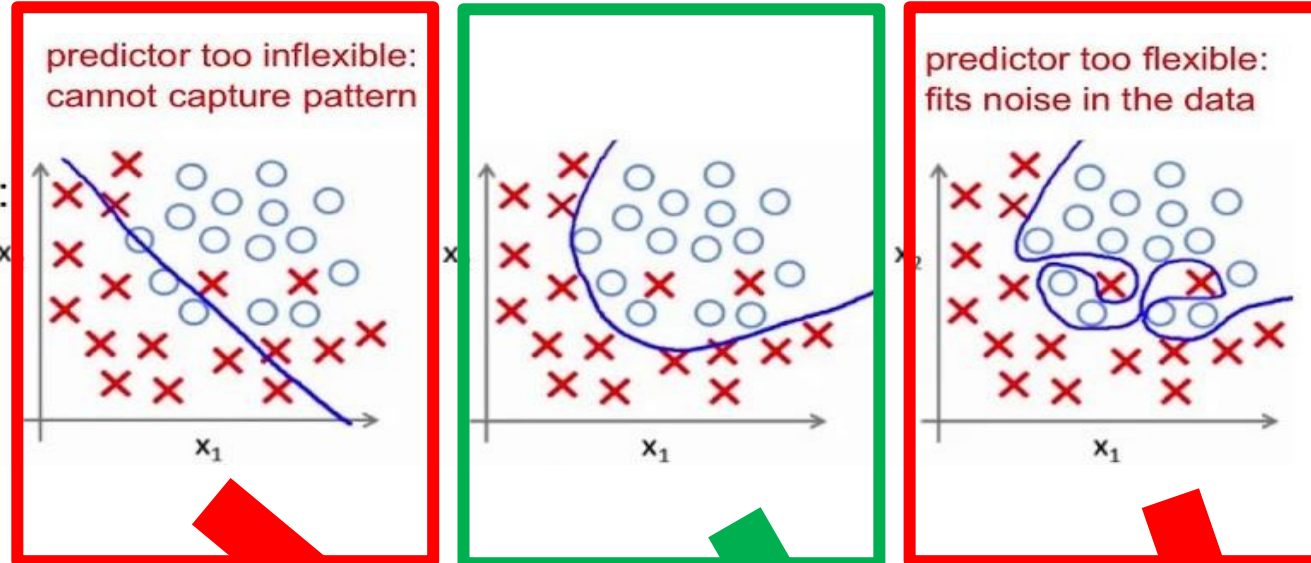
Bias-Variance Trade-Off



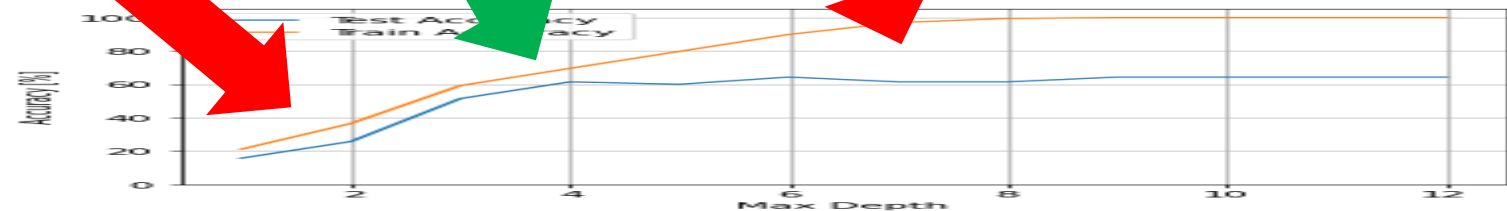
Occam's razor: The simplest model/most intuitive method is preferable

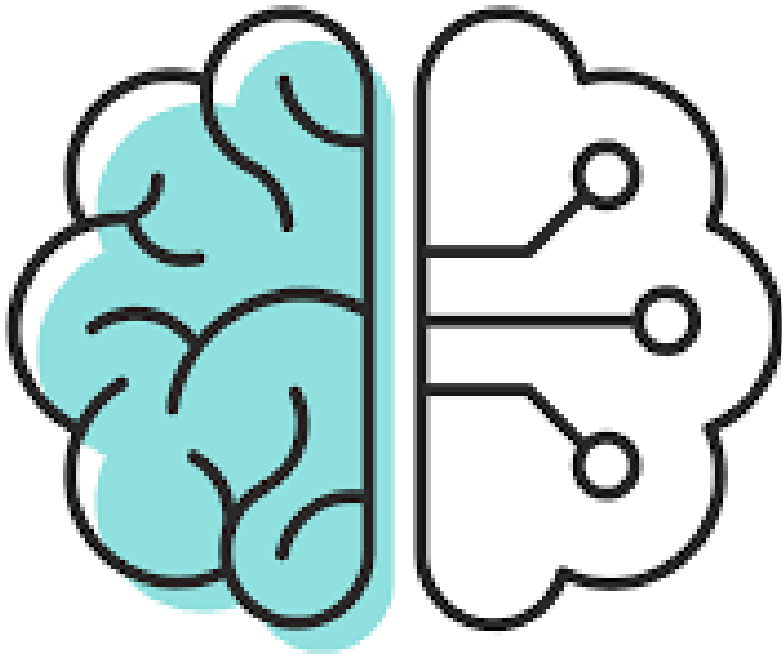
Model Selection: Under- and Overfitting for Numerical and Attribute Target Variables

Classification:



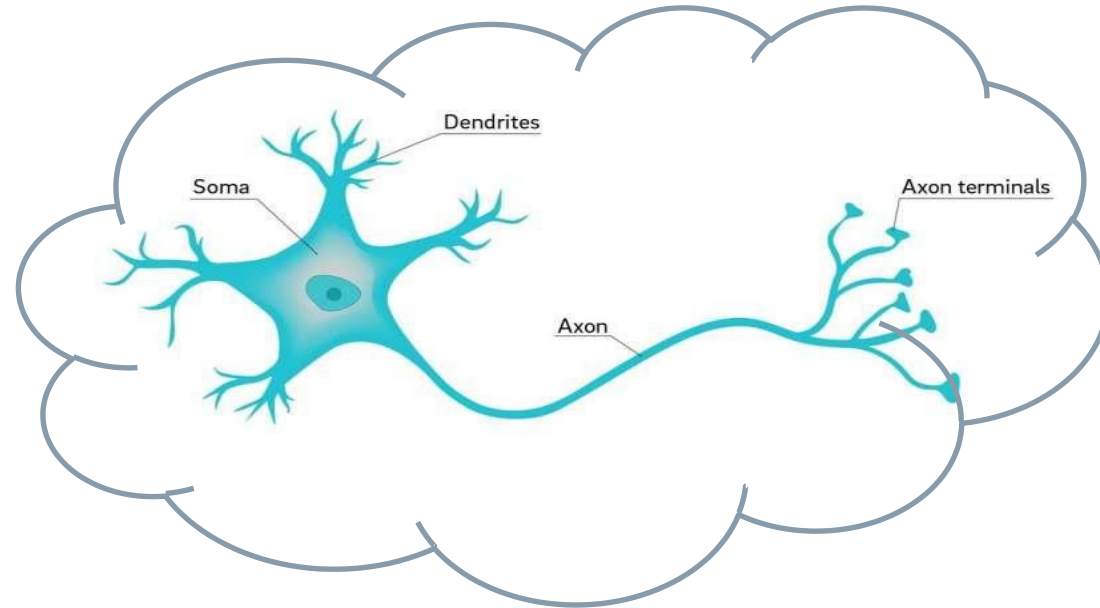
Review results from Exercise 2
Number Recognition using Decision Tree:





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Artificial Neural Nets (ANN) Are Highly Complex Models Inspired by the Biological Neural Networks that Constitute Brains

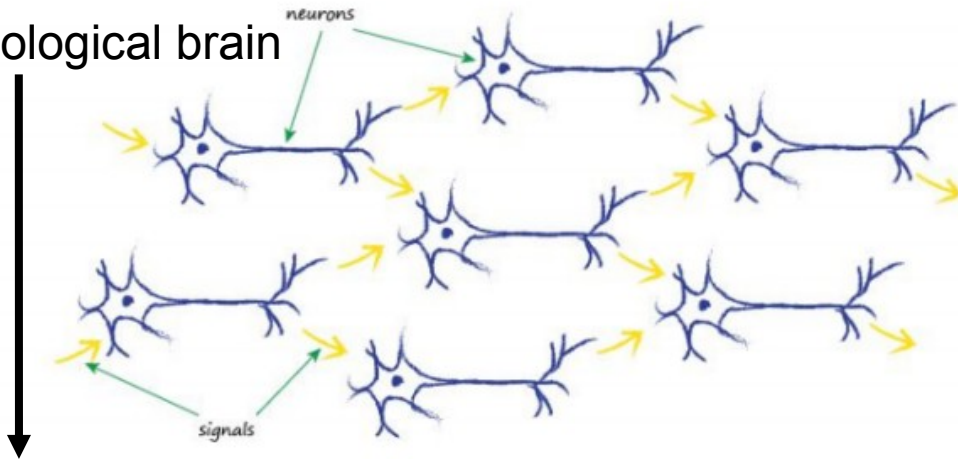


- With just ~100.000 neurons a fruit fly brain can:
 1. Fly
 2. Find food
 3. Realize and avoid danger
 4. ...
- All our computers have much more calculation capacity than 100.000 neurons
- Why is it hard for computer to do what a fruit fly can?

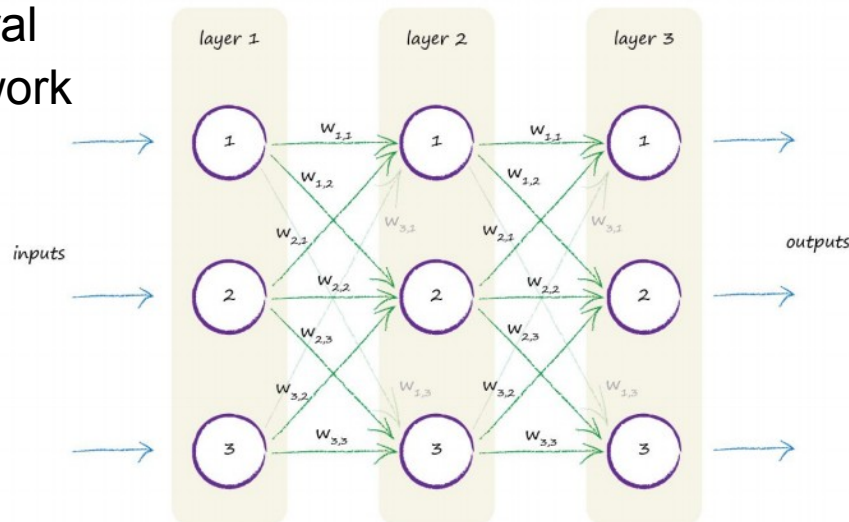
Neural Networks Take Key Inspirations from Biological Brains:

1. Take Many Inputs

Biological brain



Neural Network



1. Neurons = Variable:
 - Inputs
 - Outputs
 - Artificial Variables in between
2. Connect each neuron with all neurons to previous and next layer
3. Weight the connections between neurons:
 - Weight = slope between two variables in linear relationship
 - adjust the strength of the connections between neurons

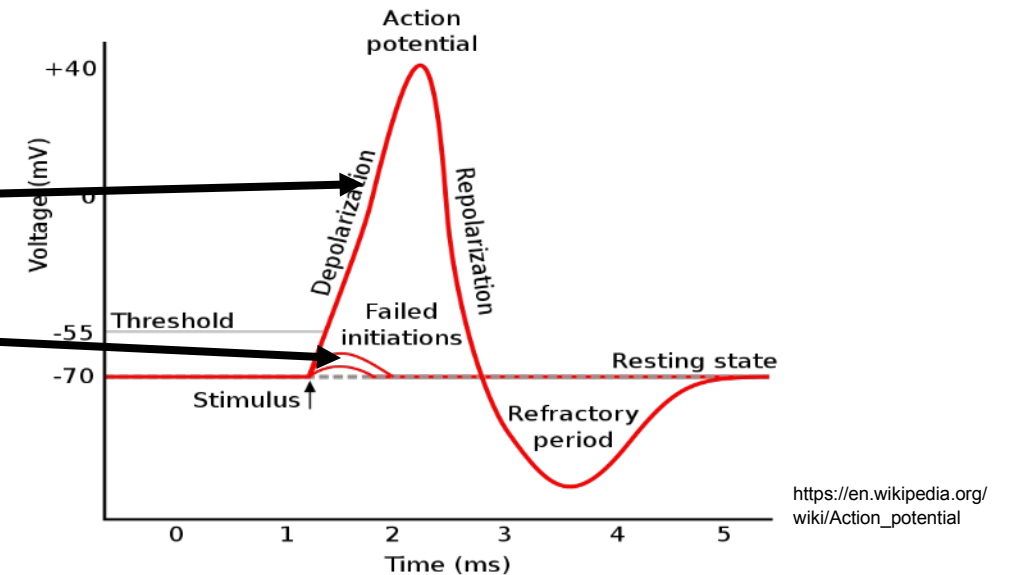
Adapted from: Tariq Rashid " Make Your Own Neural Network"

Neural Networks Take Key Inspirations from Biological Brains:

2. Use Activation Function to Differentiate btw. Important and Unimportant Inputs

Neurons work with a threshold:

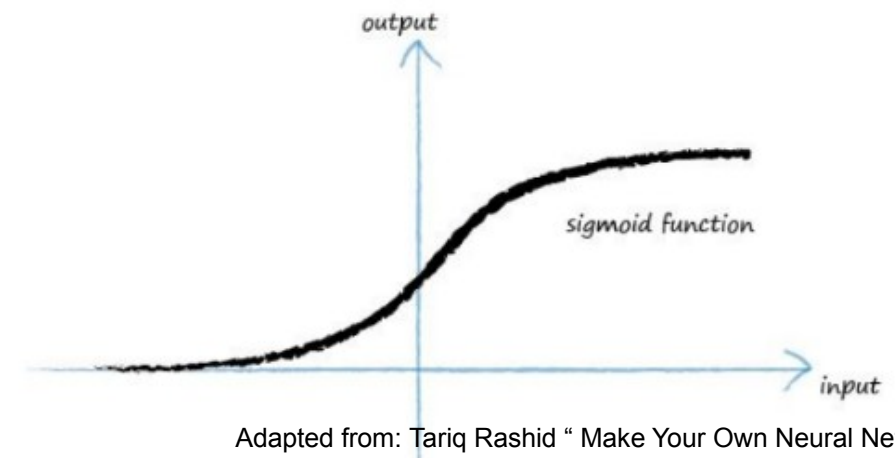
- Large inputs -> output
- Small input -> no output change



Computer use instead a “Activation Function”

- Not dealing with time
- But weighting input e.g. Sigmoid function:
 - Small changes in small or large inputs → small output change
 - Small changes in inputs near 0 → large output change

(a.n.a. Logistic function)



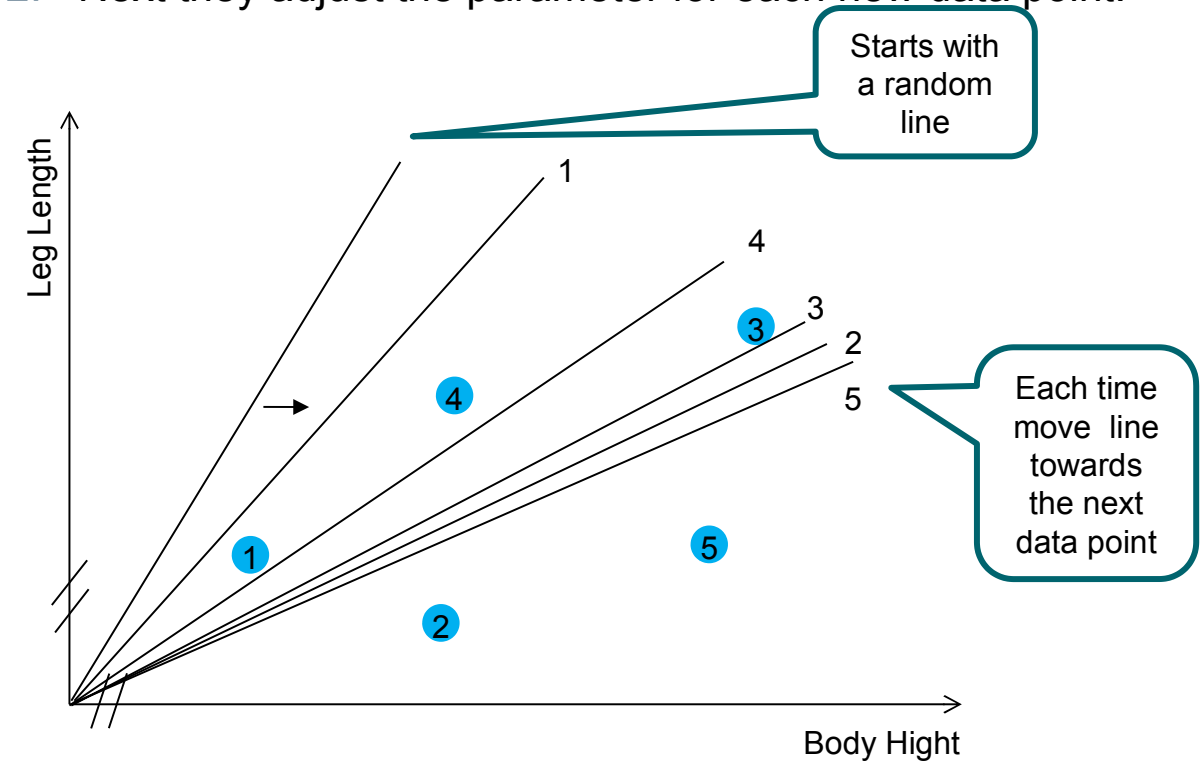
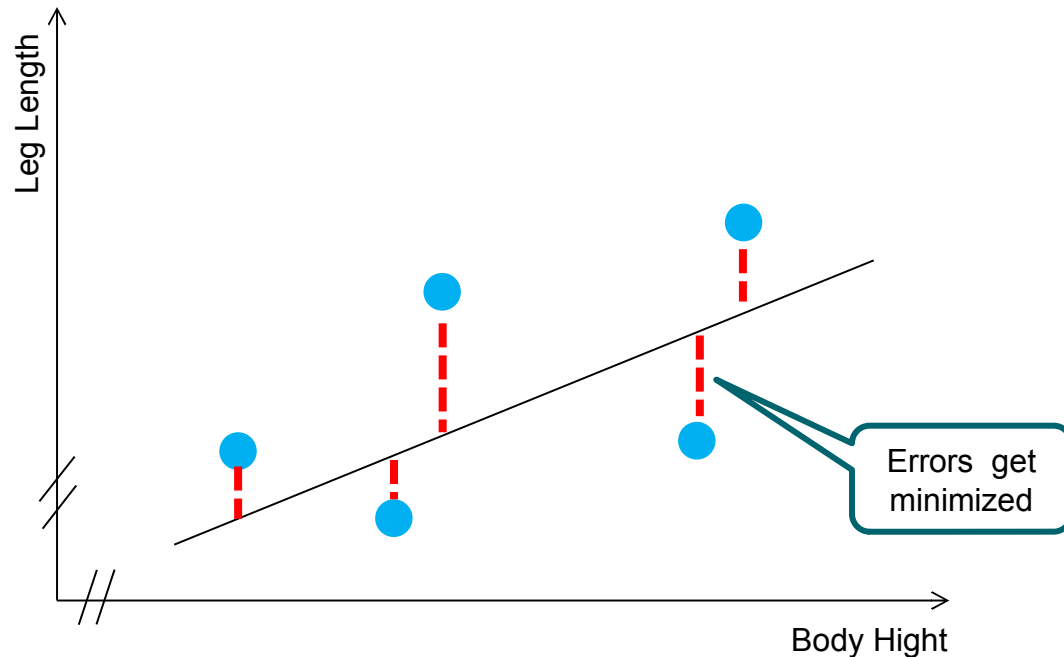
Adapted from: Tariq Rashid “ Make Your Own Neural Network”

Neural Networks Take Key Inspirations from Biological Brains:

3. Adjust model for each new data point slightly

Inference Statistics calculates a model from all data points at once

1. Neural Networks start with random parameters
2. Next they adjust the parameter for each new data point.



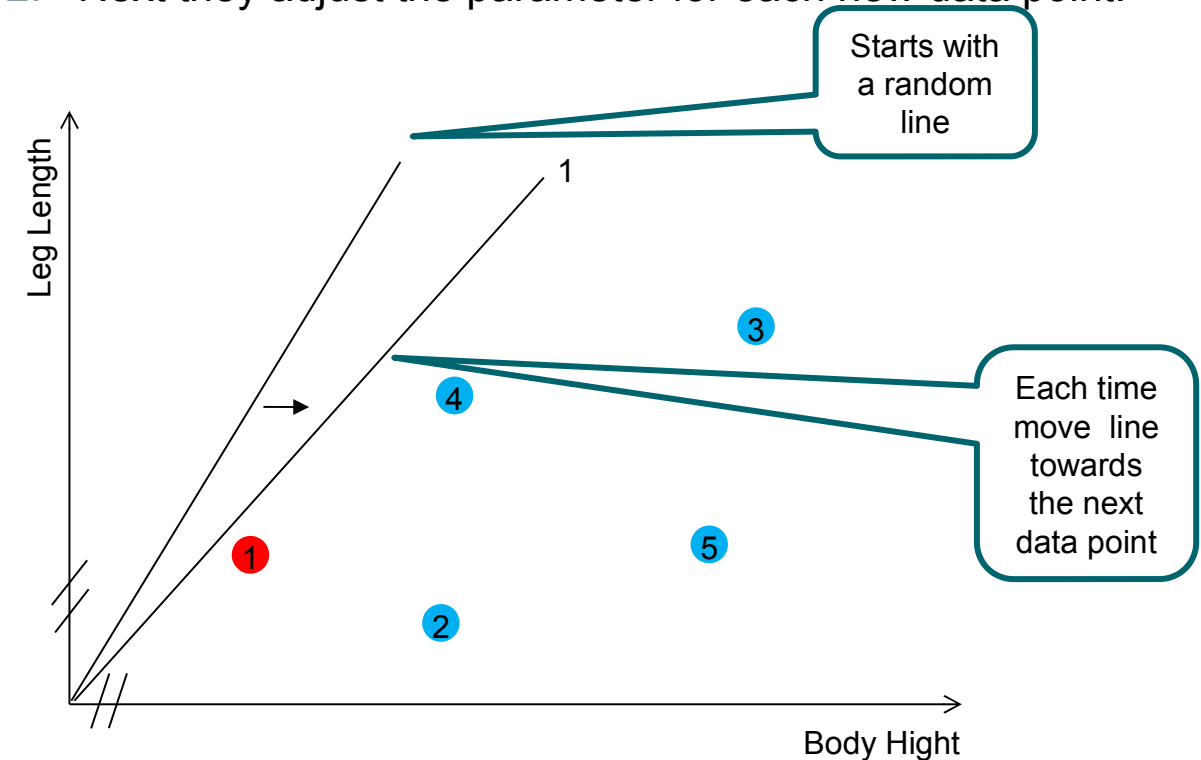
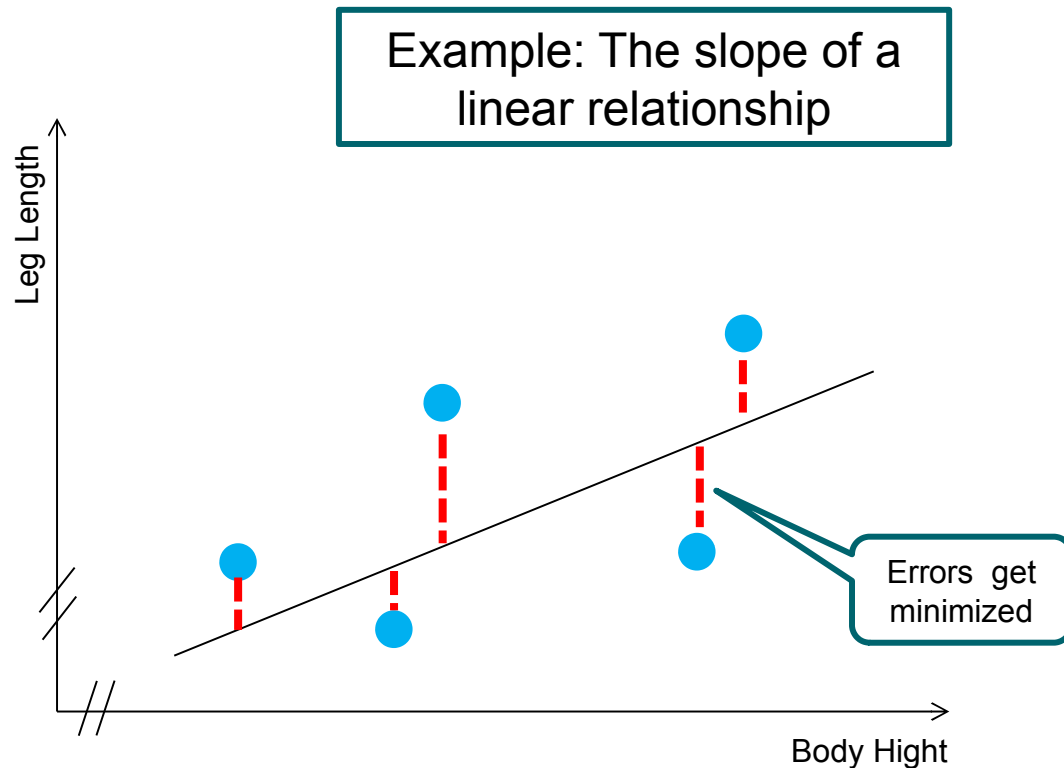
Neural Networks are much too complex to calculate them at once.

Neural Networks Take Key Inspirations from Biological Brains:

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Inference Statistics calculates a model from all data points at once

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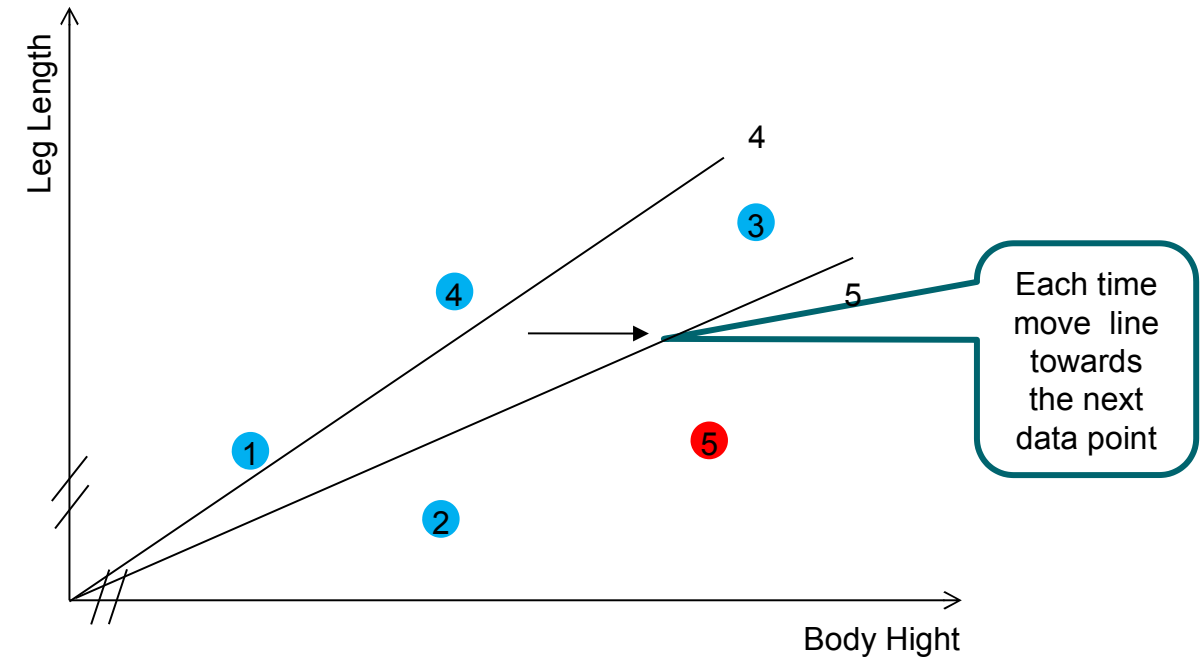
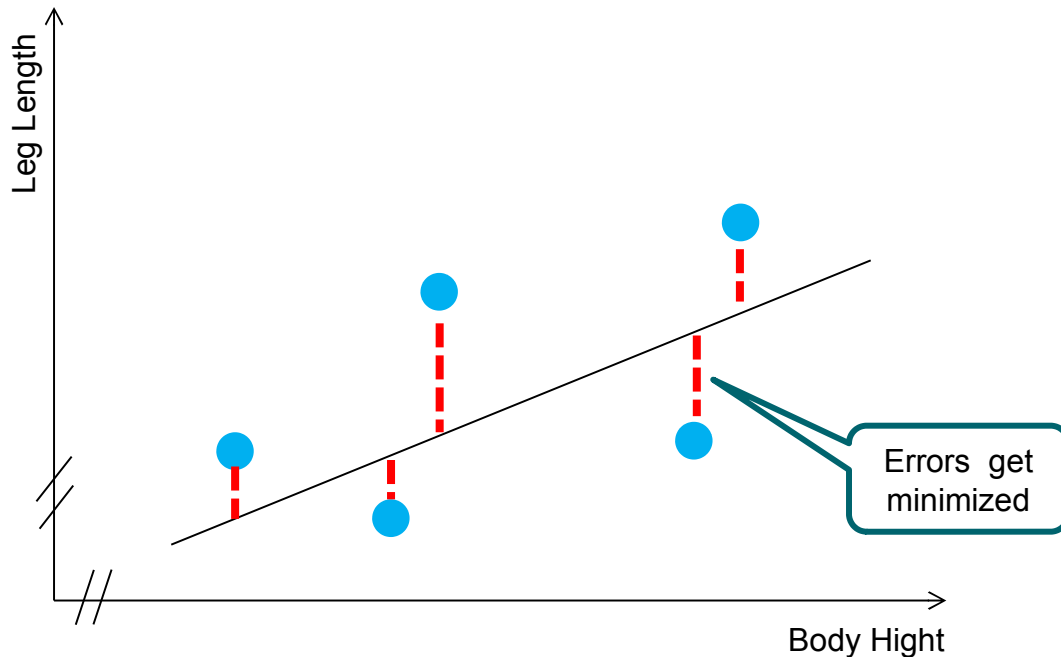


Neural Networks Take Key Inspirations from Biological Brains:

3. Adjust model for each new data point slightly

Inference Statistics calculates a model from all data points at once

1. Neural Networks start with random parameters
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Machine Learning never moves 100% onto the new data point. Why?

Why Does Machine Learning Never Move 100% onto the New Data Point?

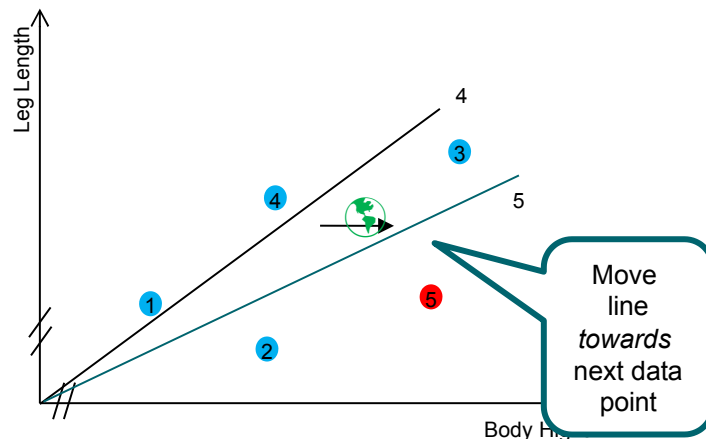
Usually, data is noisy

- Measuring errors
- Not measured factors impacting outcome
- ...

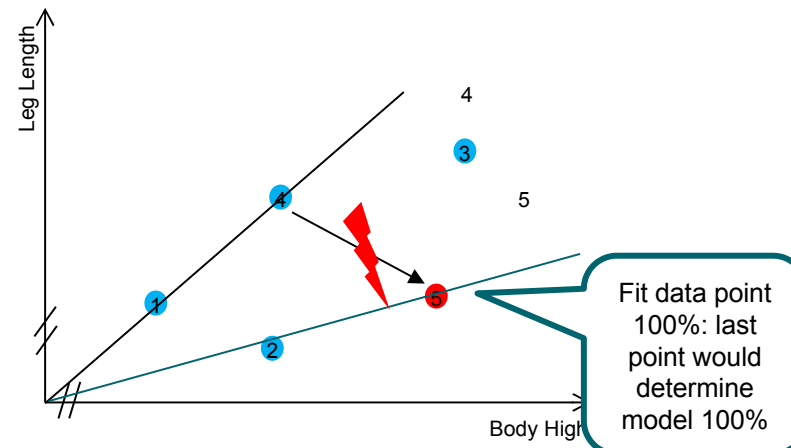
If we would fit data to 100% to the new data point, we would:

- Model jumps a lot with each data point
- Last training data point determine the entire result
- Model explains not just real effect but also noise of last training point

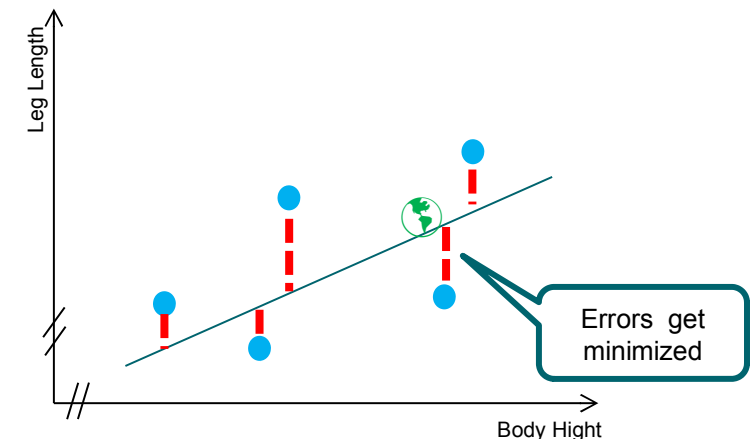
Neural Network Approach



Wrong Approach: Fit Point 100%



Trad. Inference Stats. Approach

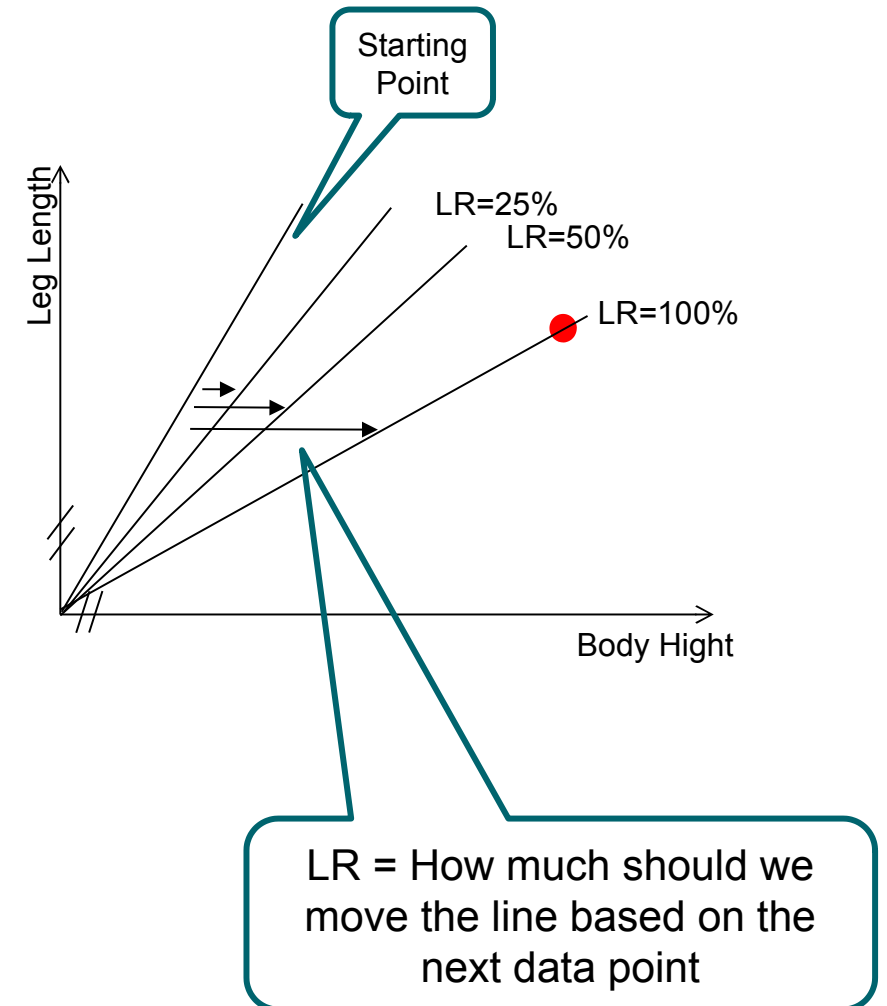


The Learning Rate (LR)

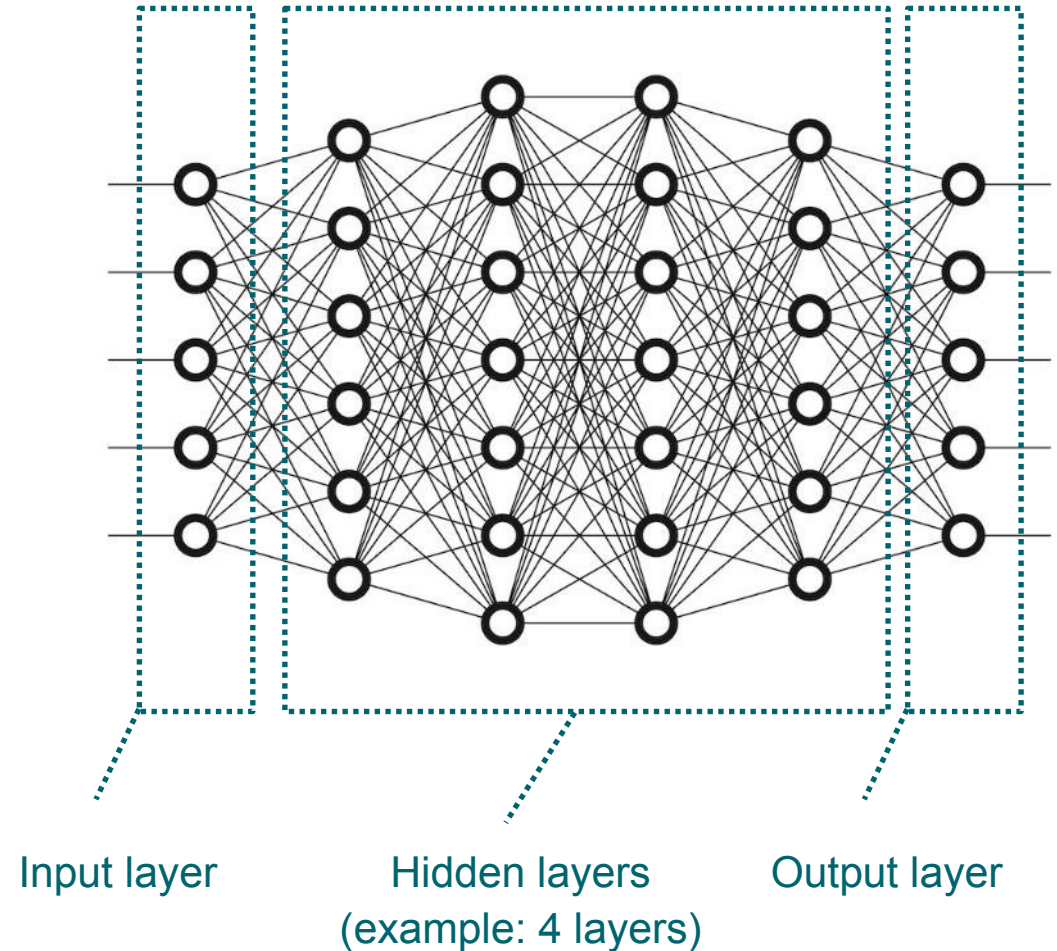
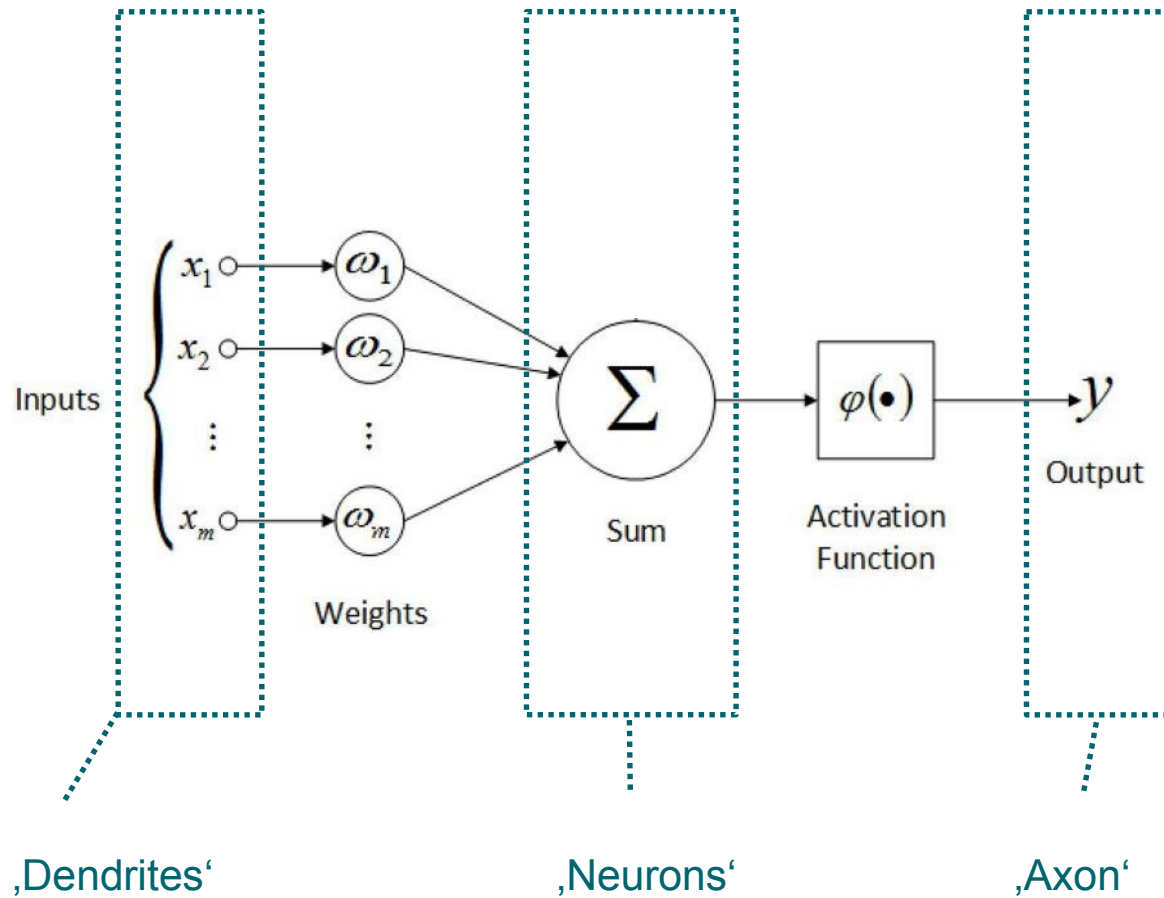
- Factor between 0 and 100%
- Determines how much the line moves to the next data point
- Dampens effect of the last data, errors and noise

Formula: $\text{Move} = \text{Learning Rate} \times \text{Distance} / 100\%$

- For big data (> Mill) often: $\text{LR} < 0.001\%$

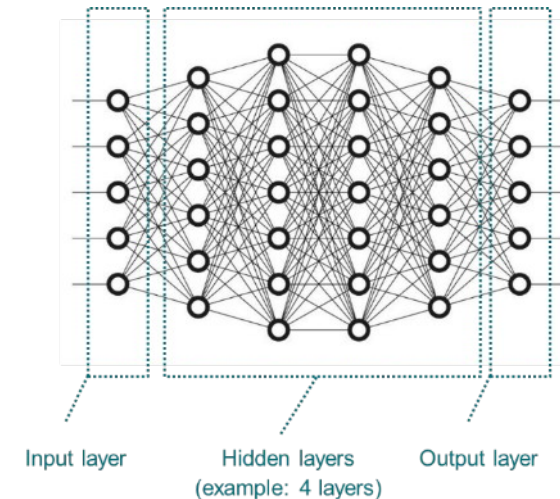
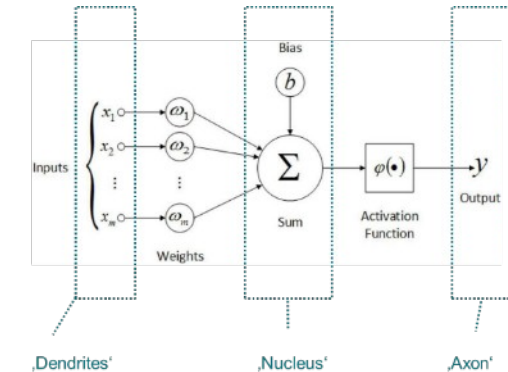


Analogies between Human and Artificial Neural Nets



Basic Steps of Training a Neural Network

- a) Select first element from the training data set
 - 1. Start with random weights (some restrictions apply) -> NN_1
 - 2. Calculate predicted output based on first element and those weights
 - 3. Calculate Error distribute to weights
 - 4. Recalculate weights based on error -> NN_2
- b) Repeat step 2-4 with all elements from the training data set
- c) Iterate through all data several times (steps a) and b)) until weights converge to one value



Traditional Regression Analysis would calculate all weights based on all sample data at once.

Exercise 3: Number Recognition Using Neural Network

1. Go to Exercise 3.1: Neural Networks
2. Find the best learning rate

Exercise 3.1: Neural Networks

Explore the Parameter **Learning Rate** with **Numbers Recognition Dataset** [↑](#)

...
...

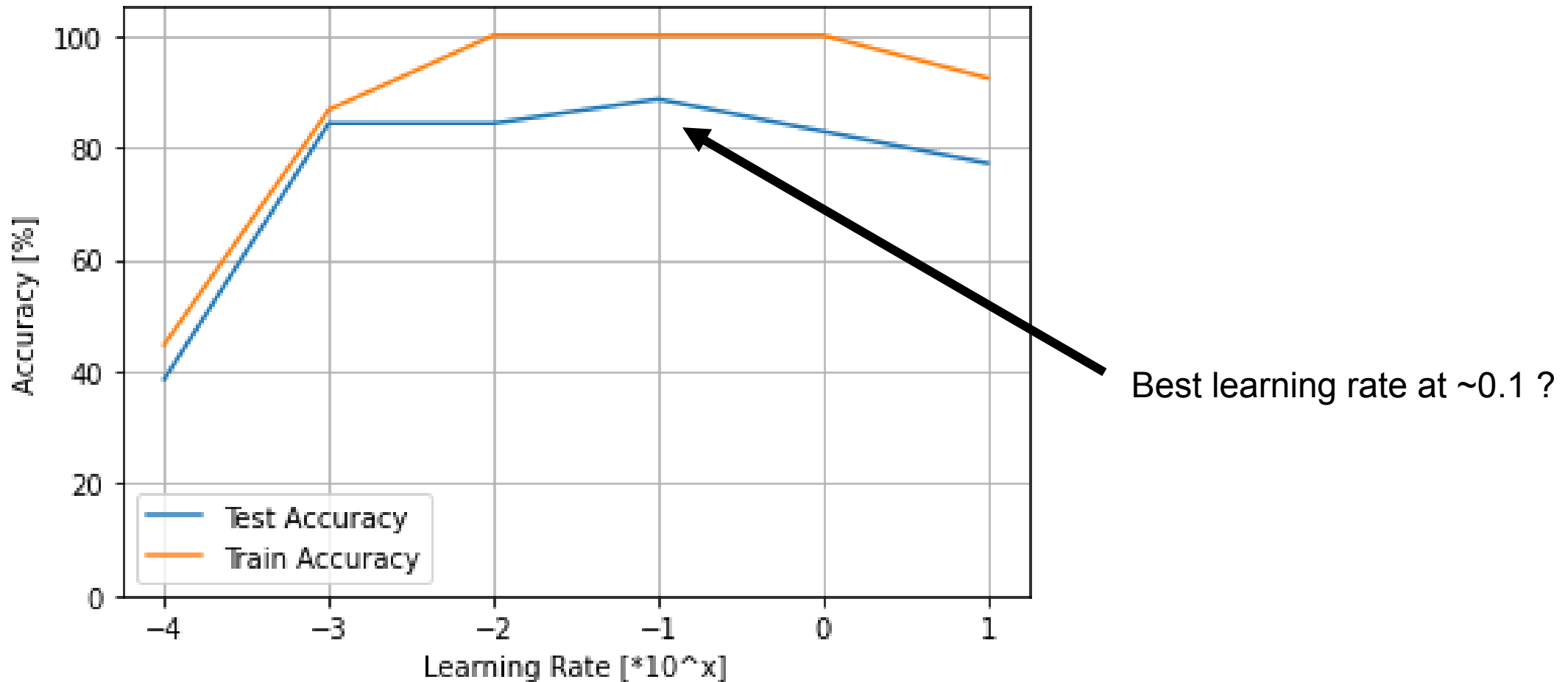


```
##### Specify parameters #####  
  
learningRate=0.0001 #Number  
  
#####  
  
#Create Neural Network  
neuralNetwork=create_neural_network(data,xData,yData,learningRate,hiddenLayers,neurons,activationFkt,  
#Save results  
resultEx31=update_result_2outputs(neuralNetwork[0],neuralNetwork[2]*100,neuralNetwork[3]*100,resultEx  
#Plot results  
logarithmic=True
```

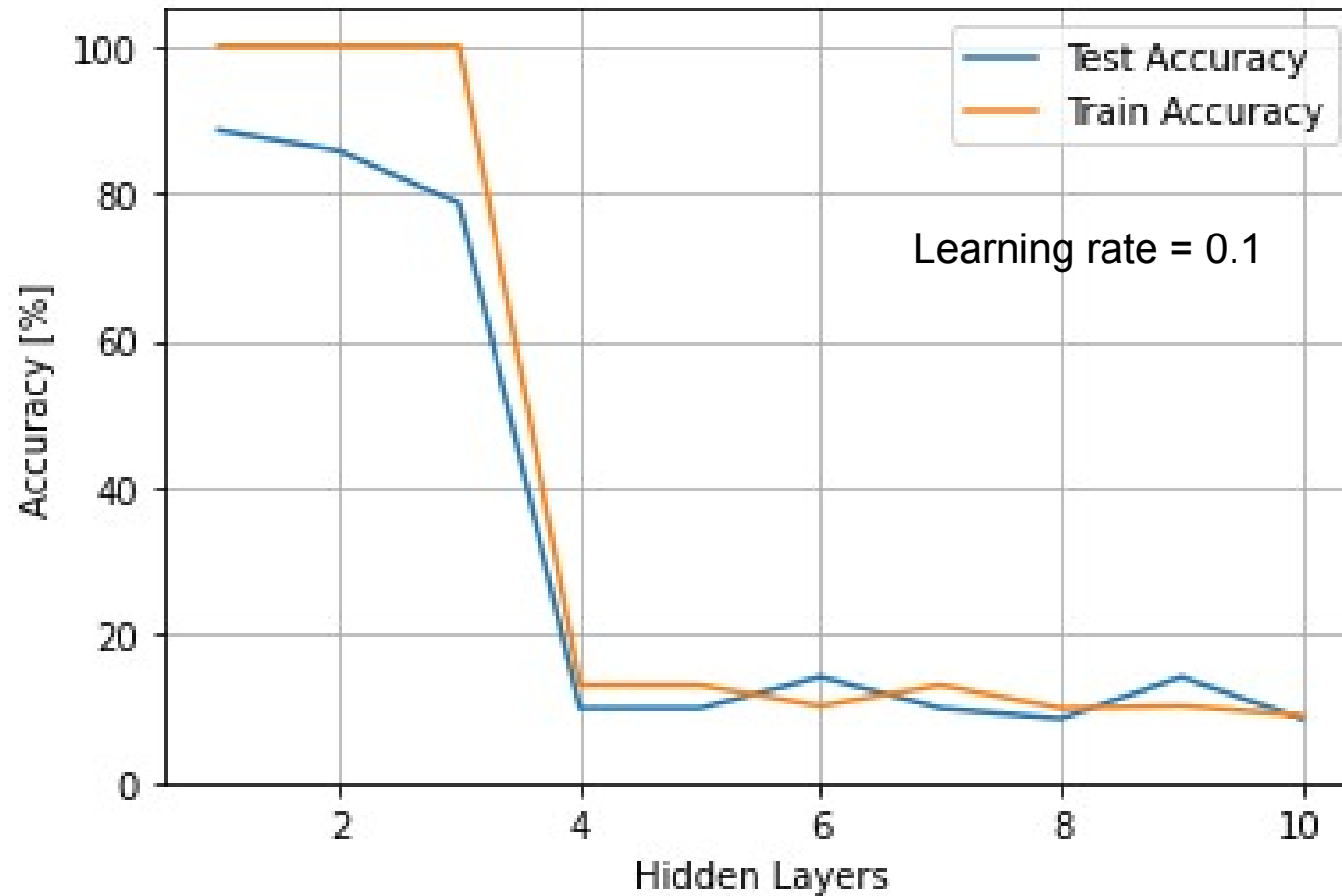
Optional

3. Go to Exercise 3.2 and find for the best learning rate the best number of hidden layers

Exercise 3: Number Recognition Using Neural Network

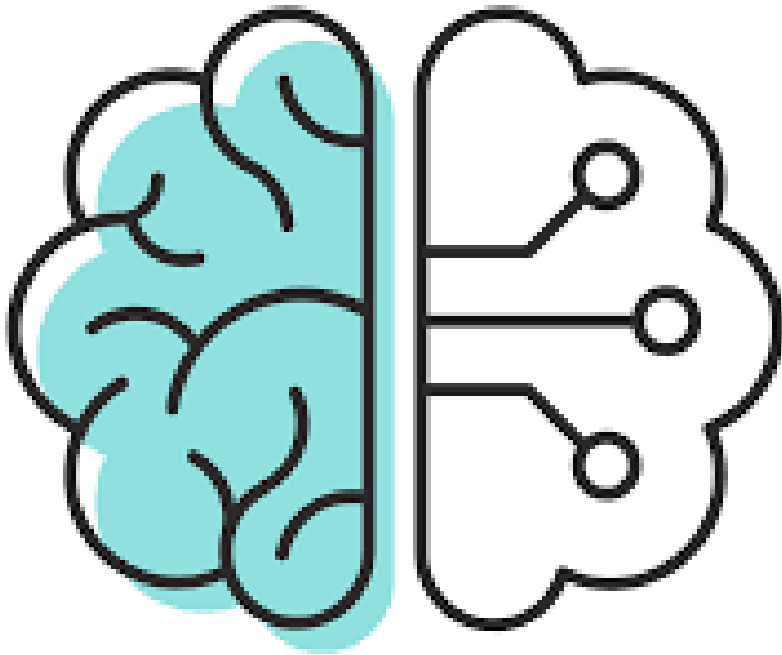


Exercise 3 Optional: Number Recognition Using Neural Network



If you increase the number of hidden layers, the model deteriorates.

For more information go to: https://scikit-learn.org/stable/modules/generated/sklearn.neural_network.MLPClassifier.html#sklearn.neural_network.MLPClassifier



- Introduction
- What Is Machine Learning?
- Decision Tree Algorithms
- Limits of Machine Learning
- Neural Networks
- Summary: Definitions and Difference between Important Approaches
- Application of Machine Learning in Our Business

Cloud Computing

- Data storage and processing in a flexible cloud rather than defined set of individual computers
- Users send just commands and receive results via internet
- Cloud provides space and calculation power on demand
- Enables processing of unprecedented amount of data

Example: powerplant data on Amazon Web Services: EnergyAI or Mosaic



Artificial Intelligence (AI)

- Anything to make computers mimic human behavior
- Includes rule based programming and machine learning
- Talked about ever since computers were invented
- Content changes with fashion/ progress

Example: Autonomous driving bus at Berlin Hospital “Charite”



<https://towardsdatascience.com/the-actual-difference-between-statistics-and-machine-learning-64b49f07ea3#:~:text=%E2%80%9CThe%20major%20difference%20between%20machine,about%20the%20relationships%20between%20variables.%E2%80%9D&text=Statistics%20is%20the%20mathematical%20study%20of%20data>

...

<https://www.ibm.com/cloud/blog/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks>

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Machine Learning vs. Conventional Statistical Models

Machine Learning

Goal:

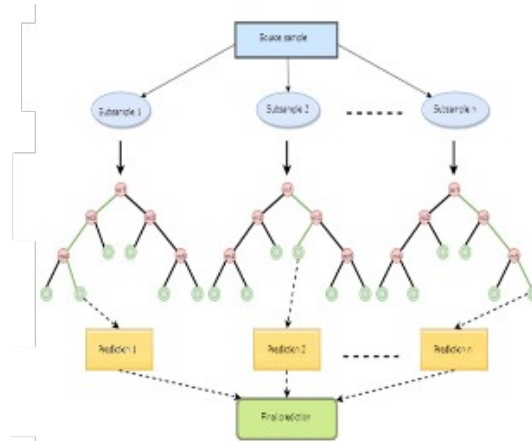
- make the most accurate predictions possible

Approach:

- Separate data in train and test data
- Train the model using training data
- Evaluate model using test data

Aspects:

- Model's predictions can often not be explained step by step ('Black Box')
- Often designed to process large data sets (big data)



Conventional Stats Modelling

Goals (depending on situation of different weight):

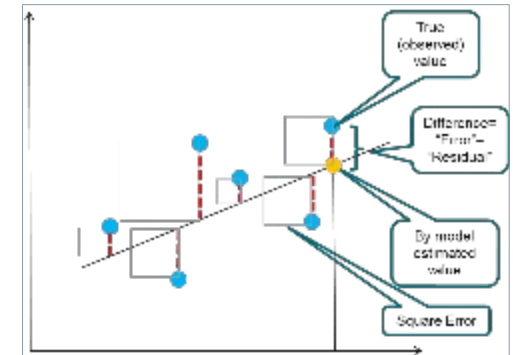
- infer something about the relationships within the data
- Stable model parameters
- be able to predict future values

Approach:

- Build model using all data
- Evaluate several aspects of the model

Aspects:

- Models predictions can usually be explained step by step (no 'Black Box')
- Can often handle less data as ML



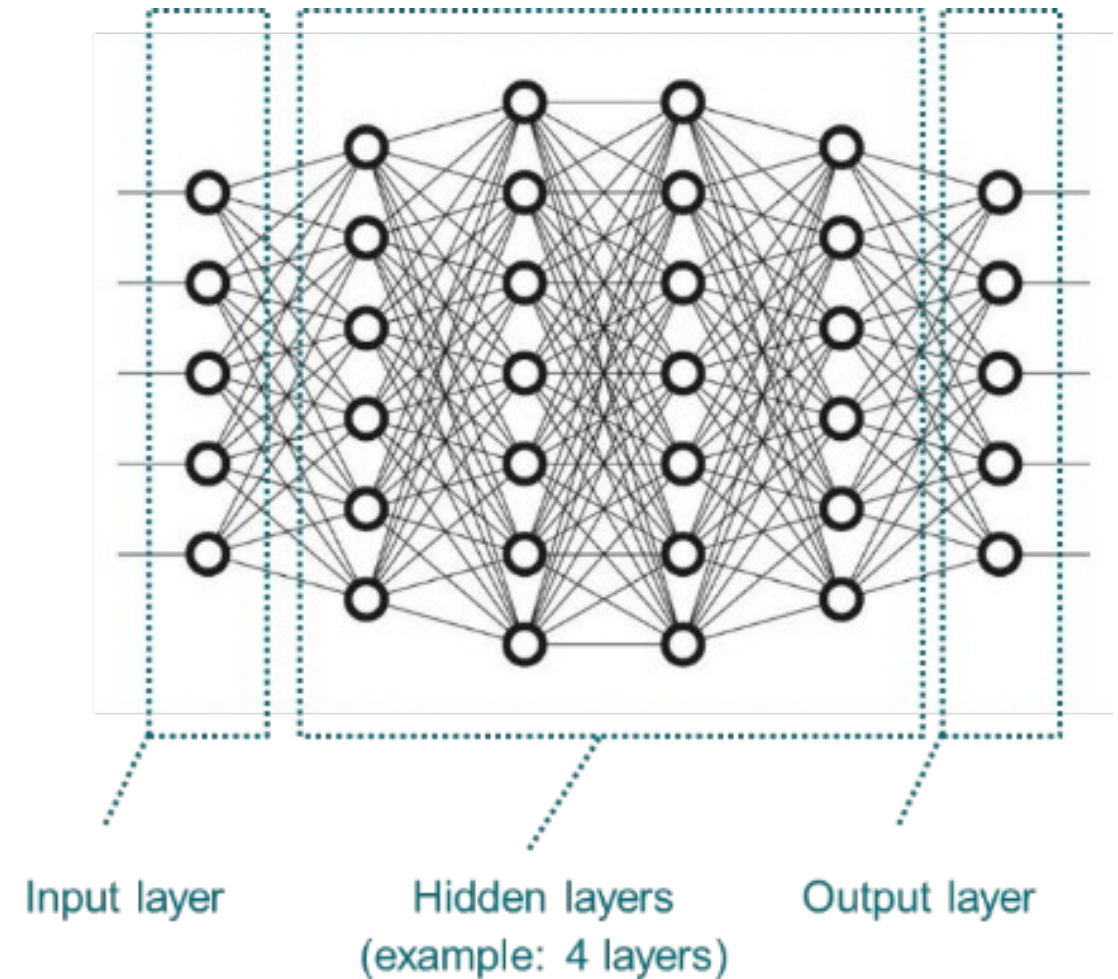
Both approaches allow computers to improve through experience and to act without explicit problem-oriented programming.

(Artificial) Neural Networks (NN, Part of ML):

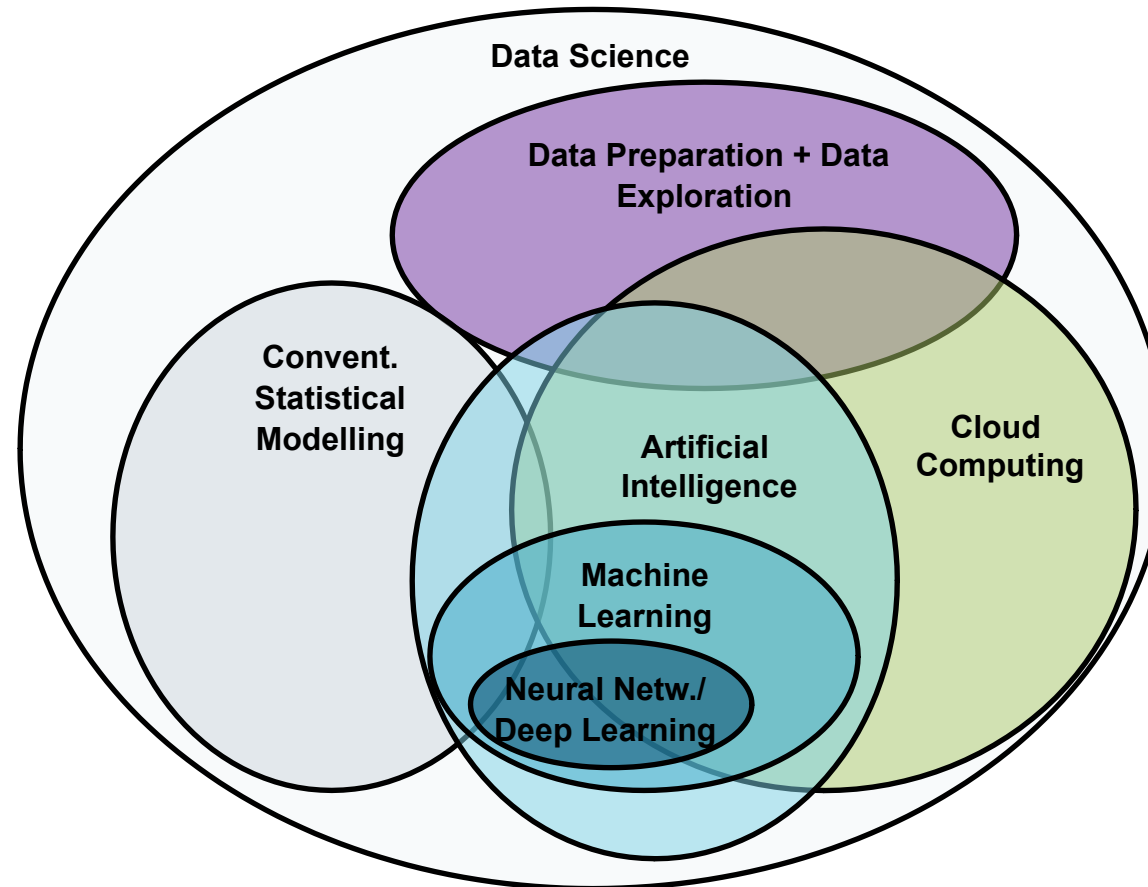
- Mimics human brains
- Inputs and Outputs are connected by “Hidden Layers”
- Hidden Layers consist of “Neurons”: little calculation models adding up inputs to an output
- Calculations take many or all inputs into account
- “Activation function” prioritizes output of each neuron
- Adopting NN by each data point, not calculating all weights at once

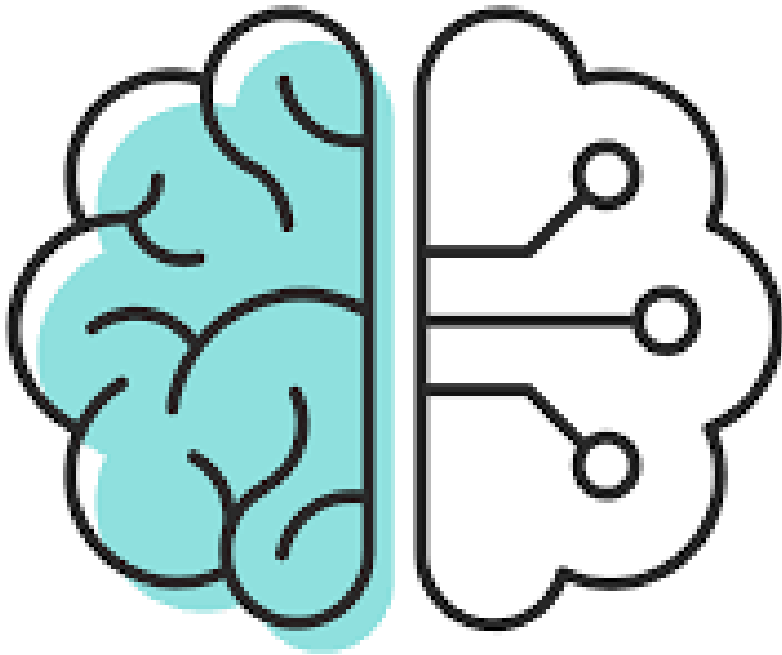
Deep Learning:

- NN with more than one hidden layer
- Examples: Speech and picture recognition



There is a lot of overlap between these concepts:





- Introduction
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Challenges of Machine Learning Application in Our Business (Review)

Consumer Data Analysis

- Little knowledge about consumer.
- Users can be tracked very closely.



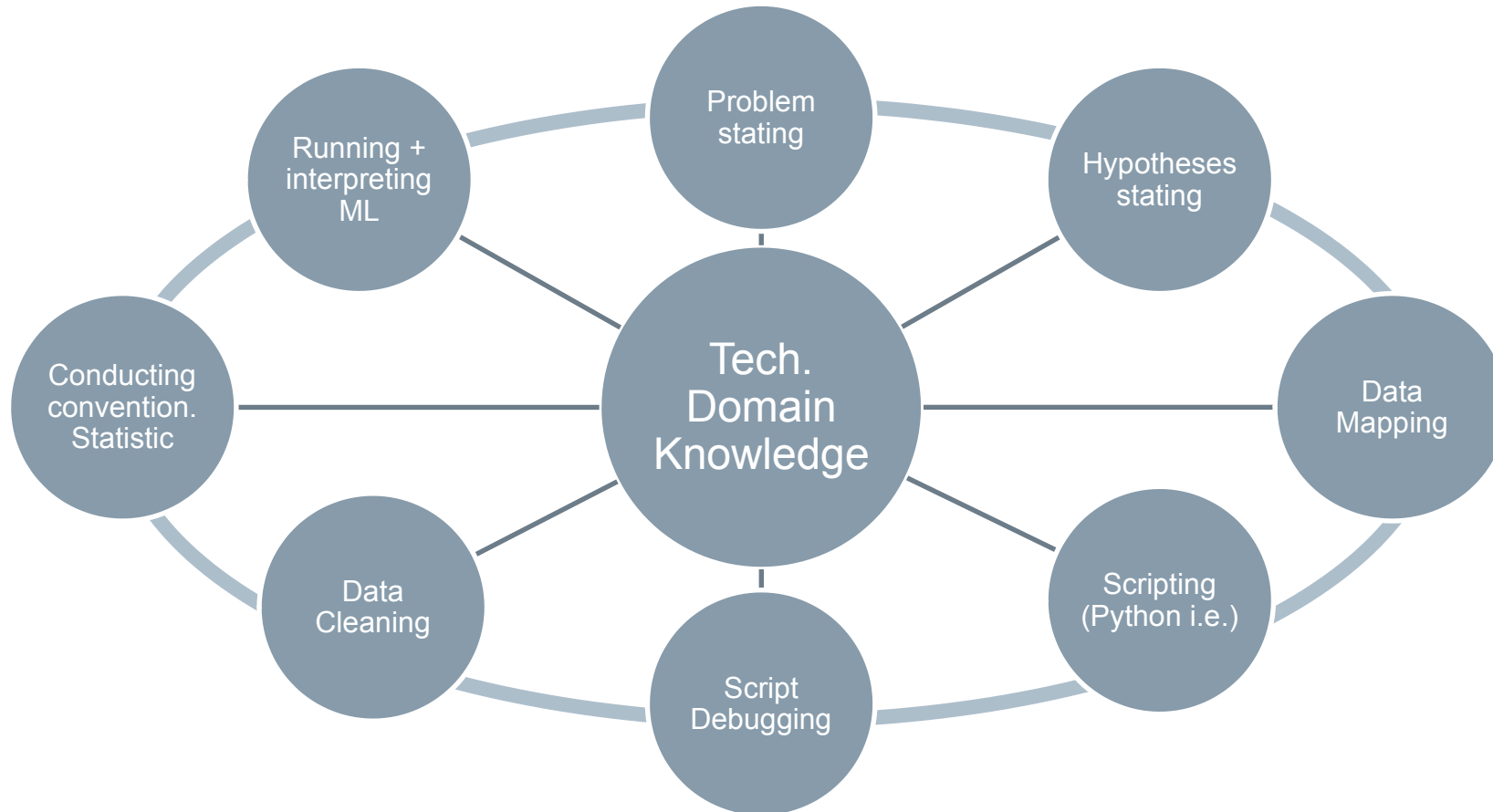
Turbine Data Analysis

- Starting from deep understanding -> very difficult to add value on top.
- Limited number of engines + change in time, operating conditions, ambient conditions.
- ML tools work best when “output data” exists (“supervised” vs. “unsupervised” learning).



We still have to learn together in which situations we can extend our knowledge with ML.

Data Science Is an Iterative Process.



For field issues differentiate between fast application and foundational development.

Overview of some Digitalization Tools and Terms

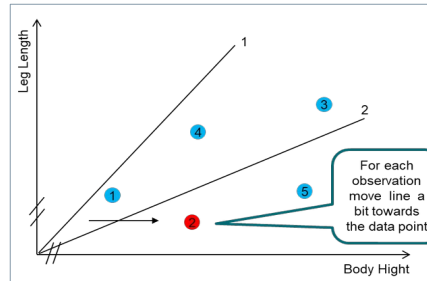
Artificial Intelligence (AI):

- Anything to make computers mimic human behavior
- Includes rule based programming and machine learning
- Talked about ever since computers were invented
- Content changes with fashion/ progress
- **Example:** Autonomous driving



Machine Learning (ML, Part of AI):

- A system that allows computers to improve through experience
- Adjusts computer model through each additional data point
- Can process larger data sets than classical inference statistics
- ML's predictions/ decisions can often not be explained step by step ('Black Box')
- **Examples:** Weather forecast

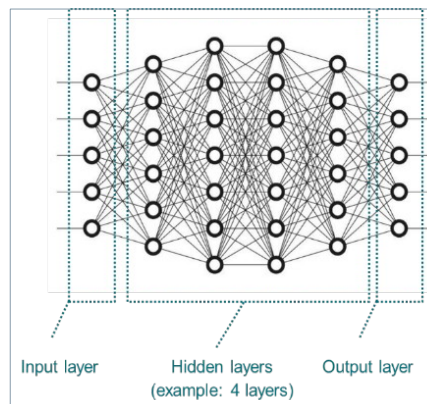


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Deep Learning: NN with more than one hidden layer

Examples: Speech and picture recognition



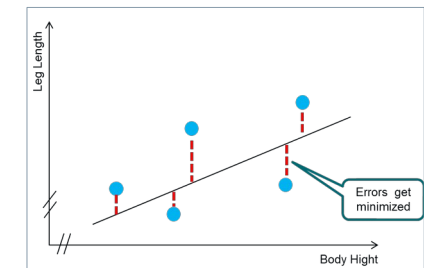
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- **Example:** powerplant data on Amazon Web Services: EnergyAI or Mosaic



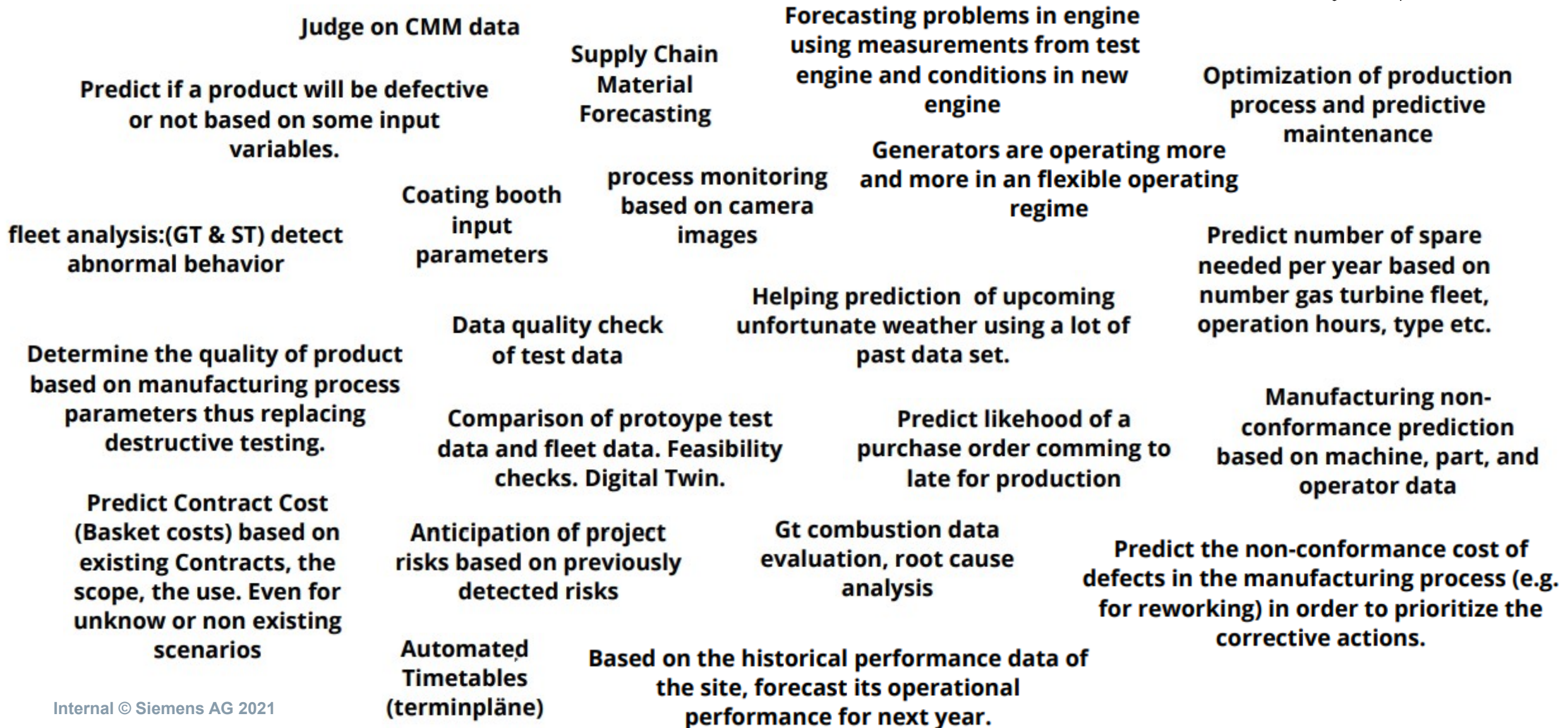
Classical Inference Statistics:

- Makes predictions based on data
- Models are calculated at once based on all data
- Updating a model with new experience requires to recalculate the entire model
(-> can not handle as much data as ML)
- Models predictions/ decisions can usually be explained step by step (no 'Black Box' as often in ML)
- **Example:** Conventional prediction of Nox-Emissions



Sources include: <https://pub.towardsai.net/differences-between-ai-and-machine-learning-and-why-it-matters-1255b182fc6>

What Could Be an Application in Your Business?



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